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Arbitrage trading between decentral and central cryptocurrency exchanges

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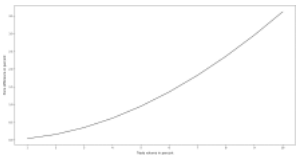
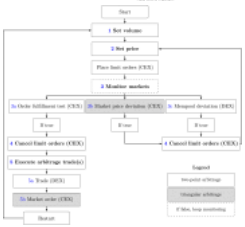
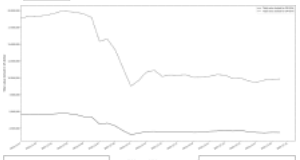
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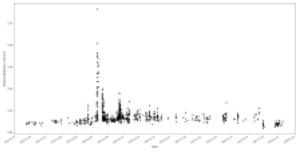
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Abstract

This paper studies liquidity fragmentation on cryptocurrency markets between automated market maker (AMM) decentralized exchanges (DEXs) and centralized limit order book exchanges (CEXs) and evaluates how practical frictions affect the execution of cross-exchange arbitrage. While most existing studies measure arbitrage opportunities using back-testing approaches based on historical price data, we analyze realized arbitrage trades executed by a high-frequency trading system across two blockchains interacting with Uniswap and Balancer type AMM protocols. Practical arbitrage differs from back-testing because it incorporates operational costs, liquidity constraints, market impact, and competition from other arbitrageurs. We develop a framework to quantify the operational costs and execution risks of cross-exchange arbitrage and examine how these frictions affect realized profitability. Four arbitrage bots deployed on two blockchains generate monthly returns of 24.91%, 33.35%, and 17.39% between November 27, 2023, and December 31, 2023, and 6.15% between September 12, 2024 and October 8, 2024 that result in an overall profit of USD 20,237.48. Our findings show that cryptocurrency markets remain fragmented between DEXs and CEXs and that capital allocation remains inefficient across exchanges. Operational costs, liquidity constraints and blockchain specific execution risks impose important limits to arbitrage. In particular, mempool uncertainty over transaction sequencing makes the effective DEX spot price uncertain prior to block confirmation, thereby reducing arbitrage efficiency between DEXs and CEXs.

JEL classification

G14
;
G15
;
G17
;
G19

Keywords

High-frequency trading
;
Distributed ledger technology
;
Smart contract
;
Arbitrage system

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1. Introduction and literature review

Cryptocurrency trading occurs on both centralized exchanges (CEXs), which operate through limit order book systems, and decentralized exchanges (DEXs), which typically rely on automated market maker (AMM) protocols. Together, these exchanges create a fragmented market structure in which identical assets are traded across central and decentral platforms that rely on different price discovery mechanisms. Arbitrage traders eliminate such price discrepancies by purchasing on the cheaper exchange and selling on the more expensive one, thereby aligning prices across markets. In cryptocurrency markets, however, arbitrage is subject to a range of operational and technological frictions that limit price integration. This paper studies fragmentation between DEXs and CEXs, documents the resulting capital inefficiencies, and examines how practical frictions affect the profitability and execution of cross-exchange arbitrage.

We provide systematic evidence on practical arbitrage by analyzing realized arbitrage trades executed by three arbitrage systems operating across DEXs and CEXs. These systems perform triangular and two-point arbitrage trades on mid market capitalization trading pairs with sufficient liquidity. The arbitrage systems exploit price differences between trading venues and generate monthly returns of 24.91%, 33.35%, and 17.39% between November 27, 2023, and December 31, 2023, and 6.15% between September 12, 2024, and October 8, 2024, amounting to an overall profit of USD 20,237.48. Our analysis demonstrates how operational costs, liquidity constraints, and blockchain specific execution risks limit arbitrage. In particular, mempool uncertainty regarding transaction sequencing makes the effective DEX spot price uncertain prior to block confirmation and reduces arbitrage efficiency between DEXs and CEXs.

According to DefiLlama,[1] more than 250 blockchains operate simultaneously, and each blockchain offers different security models, consensus mechanisms, and performance characteristics. Weak connectivity and interoperability between these networks create a fragmented capital system in which liquidity isolates across multiple chains and trading venues. Caldarelli (2021) provides an overview of wrapped token issuing procedures where users can transfer assets across blockchains through bridging provider that lock an asset on its native chain and mint a wrapped representation on another chain. Although bridging improves connectivity between blockchains, it introduces additional costs and operational frictions. Alternatively, to wrapping providers, CEXs help to facilitate liquidity transfers between blockchains by allowing users to deposit assets from one chain, trade them within the exchange environment, and withdraw them to another blockchain. Next to CEXs, DEXs rely on smart contracts deployed directly on the blockchain to facilitate peer to peer trading without intermediaries as defined by Jiao et al. (2024). Most DEXs rely on AMM models rather than limit order book mechanisms commonly used in CEXs.[2]

A growing literature examines arbitrage opportunities in cryptocurrency markets using historical price data. Bruzgė and Šapkauskienė (2022) analyze Bitcoin arbitrage opportunities across 13 CEXs using minute level data from January 2018 to April 2020 and find that price discrepancies persist despite the rapid growth of cryptocurrency markets. Borri and Shakhnov (2022) study Bitcoin price disparities across 29 CEXs and show that arbitrage opportunities exist but are not riskless once transaction costs and trading constraints enter the analysis. Makarov and Schoar (2020) document substantial price differences across geographically segmented cryptocurrency markets and report average spreads of up to 15 percent between Korean and U.S. exchanges during the 2017–2018 period. Crépellière et al. (2023) extend this line of research and show that cross-exchange price deviations decline over time but persist even in more mature cryptocurrency markets.

Although, this literature provides important evidence on arbitrage across CEXs, most studies rely on back-testing approaches based on historical price data. These approaches assume that trades execute at observed prices and, therefore, neglect key frictions that arise in practical arbitrage trading, including slippage, liquidity constraints, competition among arbitrageurs and market impact. In addition, the literature rarely examines arbitrage between AMM DEXs and limit order book CEXs, partly because collecting high-frequency blockchain trading data and linking it to realized trades requires substantial technical effort.

Our research addresses these limitations by analyzing realized arbitrage trades executed between DEXs and CEXs. Instead of relying on historical price data, we analyze arbitrage transactions that generate realized profits in cryptocurrency markets. This approach allows us to compare theoretical arbitrage spreads with realized arbitrage outcomes and to quantify how operational trading costs, liquidity constraints, and execution risks limit arbitrage profitability. The results highlight the difference between measurable arbitrage spreads derived from historical price data and realized arbitrage spreads from actual trading activities that incorporate operational costs and market frictions.

To identify and analyze realized arbitrage opportunities, we deploy a high-frequency trading system that executes arbitrage trades between DEXs and CEXs. We focus on DEX to CEX arbitrage because innovations such as smart order routing and DEX aggregators reduce arbitrage opportunities between DEXs operating on the same blockchain, while professional market makers increasingly dominate arbitrage between CEXs. Consistent with the limits to arbitrage framework of Schleifer and Vishny (1997), our results show that specialized participants conduct most arbitrage trading in cryptocurrency markets because successful execution requires technical infrastructure and market expertise.

The remainder of the paper proceeds as follows. Section 2 develops the mathematical framework for slippage and transaction costs in cross-exchange arbitrage. Section 3 describes the arbitrage algorithm and trading system. Section 4 presents empirical results from the deployed arbitrage systems. Section 5 analyzes the performance of the trading systems across blockchain environments. Section 6 discusses the determinants of arbitrage profitability and the implications for market efficiency. Section 7 examines how evolving market structures affect future arbitrage opportunities. Section 8 concludes the paper.

2. Mathematical foundations

2.1. Slippage

The volume of a trade interacting with an AMM DEX directly influences the spot price. Consequently, arbitrage traders must assess the anticipated price impact of their orders, quantified as slippage. Arbitrage traders incorporate the expected cost of slippage into their quotation decisions. Xu et al. (2023) define slippage as the difference between the spot price at the initiation of a trade and the actual price realized upon its execution.[3] The following demonstrates how to compute slippage for a DEX order and a market order on a CEX. The DEX AMM conservation function enables the deterministic calculation of expected slippage. Calculating slippage at a DEX is contingent upon the protocol's underlying conservation function. Xu et al. (2023) provide a comprehensive overview of various AMM protocols, detailing each protocol's conservation function, spot exchange rate, slippage, and divergence loss. Depending on the exchange's conservation function, this systematic framework assists investors in identifying embedded risks and predicting expected trade outcomes.

To maintain simplicity, this paper adopts the variables presented by Xu et al. (2023) to demonstrate the conservation function, exchange rate, and slippage calculation at the DEX. The DeFiChain DEX is a Uniswap V2 model, and correspondingly, its conservation function is defined as[4]: (1)$K=r_1 \cdot r_2$where K denotes the invariant, and r_1 and r_2 denote the reserves for asset 1 and asset 2, respectively. At each point in time t, the spot exchange rate, E_2, is given by: (2)$E_2=r_1/r_2$When a trader interacts with the pool to exchange assets, they provide an amount x_1 to the pool to receive an amount x_2. The expected slippage for this trade, $s\text{DEX}$, is[5]: (3)$s\text{DEX}=(x_1/x_2)/(r_1/r_2)-1=x_1/r_1$which represents the ratio of the actual price achieved upon trade execution to the spot exchange rate.

The order matching process in a limit order book system operates differently from that of the AMM model, which changes the computation of the expected slippage at CEXs. For limit orders, there is no slippage since they are executed at a fixed price. However, market orders can exhibit slippage costs. Therefore, arbitrage traders determine the expected slippage by fetching current quotes from the limit order book. For a market order, the realized price $P\text{realized}(x)$ is computed as follows: (4)$P\text{realized}= \sum_{i=1}^{n} i=1(x_i \cdot P_i)/x$where x_i and P_i represent the trading volume and price for quote i, respectively, and x denotes the total trading volume at the CEX, calculated as $\sum_{ni=1} x_i$. Consequently, the slippage at the CEX, $s\text{CEX}$, is determined by: (5)$s\text{CEX}=P\text{realized}/P\text{spot}-1$where $P\text{spot}$ denotes the spot price at the CEX. Calculating expected slippage in a limit order book exchange is particularly challenging and prone to inaccuracies for two reasons. First, high-frequency traders frequently adjust their quoted prices P_i and volumes x_i, resulting in greater liquidity volatility in the order book compared to the more stable, long-term liquidity provision at DEXs, which change less frequently. Second, exchanges such as KuCoin[6] allow orders to be hidden within the order book. This can result in an overestimation of the arbitrage trader's price impact, as the actual liquidity in the order book is greater than what is being measured. KuCoin offers a hidden order feature that allows traders to submit limit orders which are invisible to other market participants. Arbitrage traders and market makers compete for the top positions in the order book to maximize the likelihood of their orders being filled. This competition often results in price auctions, where bots continuously adjust their bids and asks to outcompete one another. The hidden order functionality helps to mitigate these resource-intensive auctions by enabling traders to participate without publicly disclosing their quotes. Note, not all CEXs offer hidden orders which is why price auctions remain visible in limit order book systems.

Many researchers conducting back-testing of arbitrage profits use historical datasets and overlook hidden costs, such as slippage. However, incorporating these costs is essential, as they can significantly affect the accuracy of profit estimates. Neglecting slippage can lead to overestimated arbitrage opportunities. Accounting for slippage and other transaction costs provides a more realistic evaluation of profitability in practical trading environments.

2.2. Profit and cost function

The slippage and trading costs reduce the realized profit from an arbitrage trade. The costs $c\text{trade}$ for a two-point and triangular arbitrage trade include all fees and slippages from the CEX and DEX. The following cost model assumes that a two-point arbitrage trade involves one limit order at the limit order book at the CEX and one order at the AMM DEX. For triangular arbitrage, the model assumes a limit order and a market order at the CEX which mimics one trade at the DEX. The trade cost is a variable cost and depends on the volume traded at the DEX and the volume traded at the CEX. For two-point arbitrage, the trade cost $c_{two-point}\text{trade}$ is calculated as: (6)$c_{two-point}\text{trade}=f\text{CEX}_{\text{limit}}+f\text{DEX}+s\text{DEX}$where $f\text{CEX}_{\text{limit}}$ denotes the fee for placing a limit order at the CEX and f_{DEX} denotes the fee for a trade at the DEX. For triangular arbitrage, the trade cost $c_{triangular}\text{trade}$ additionally includes the fee for placing a market order $f\text{CEX}_{\text{market}}$ and the slippage $s\text{CEX}$ at the CEX, and is calculated as: (7)$c_{triangular}\text{trade}=f\text{CEX}_{\text{limit}}+f\text{CEX}_{\text{market}}+f\text{DEX}+s\text{CEX}+s\text{DEX}$[7] The profit function π for two-point and triangular arbitrage trades is: (8)$\pi=$

$$\underbrace{(\rho - c\text{trade} - \Delta p\text{DEX} - \Delta p\text{CEX})}_{\text{realized spread}} \cdot x - F\text{transaction}$$

where ρ denotes the configured cross-exchange arbitrage spread, $\Delta p\text{CEX}$ and $\Delta p\text{DEX}$ indicate the price change in basis points at the CEX and DEX between quotation and execution, and $F\text{transaction}$ denotes the fixed fee for every transaction on the blockchain. $F\text{transaction}$ varies among blockchains, e.g., before the introduction of Layer 2 solutions, it often exceeded 10 US dollars on the Ethereum main net, while $F\text{transaction}$ on Solana rarely exceeds 10 cents. Note that for most CEXs, $f\text{CEX}_{\text{limit}} < f\text{CEX}_{\text{market}}$, since limit orders provide liquidity to the order book, while market orders take liquidity. Exchanges aim to increase liquidity by incentivizing liquidity providers. At KuCoin,[8] traders exceeding a 30-day trading volume of 350 million USDT do not pay market-making fees, resulting in $f\text{CEX}_{\text{limit}}=0$. Besides trading costs, an additional cost occurs due to liquidity rotation. $C\text{travel}$ represents the cost incurred when circulating liquidity between exchanges and is calculated as follows: (9)$C\text{travel}=F\text{withdrawal}+F\text{wrapping}$where $F\text{withdrawal}$ denotes the withdrawal cost from the CEX, and $F\text{wrapping}$ represents the wrapping fees through the wrapping provider. A reduced cost function allows a decrease in ρ while maintaining the same profit level, facilitating earlier execution of the limit order. Consequently, a reduction in ρ leads to a narrowing of the cross-exchange spread between the two exchanges. It shows that the cross exchange arbitrage spread is heavily dependent on the cost structure at and between both markets. If the exposure position of the arbitrage portfolio runs low at one exchange, it requires to circulate funds, leading to costs of funds that travel across exchanges $C\text{travel}$. For the majority of CEXs, $F\text{withdrawal}$ is a fixed fee that occurs whenever the funds are withdrawn from the exchange. Additionally, the circulation of funds may demand to use a wrapping service, where the wrapping fee $F\text{wrapping}$ adds to travel costs.[9] The results in Section 4 show how costly the rotation of liquidity can be. One way to reduce travel costs is the reduction of the desired spread ρ for one order, either bid or ask limit order, when the bot's liquidity runs low at either exchange. The reduced

spread level increases the probability of trade executions in one direction and, correspondingly, leads to the autorebalancing effect of the exposure position. The trading system on the SEI blockchain in Section 5 advances our algorithm and incorporates the autorebalancing effect. The flexible spread management decreases C_{travel} but comes at a trade-off for a reduced realized profit or even unprofitable trades. Another option to reduce C_{travel} is to increase the exposure position of the arbitrage portfolio. As $F_{\text{withdrawal}}$ and F_{wrapping} tend to be fixed fees (dependent on arbitrage environment), arbitrage traders can reduce travel costs C_{travel} with an increased exposure portfolio that allows to reduce the frequency of circulations. This allows an increase in the absolute profit but comes at the cost of a lower rate of return due to the increased portfolio position.
The cost functions c_{trade} and C_{travel} are dependent on the exchanges, trading pairs, wrapping providers, and blockchains. A single cost element can be influential for the entire strategy of the algorithm. For example, the Ethereum blockchain has comparable high fees for transactions $F_{\text{transaction}}$ that varied between June 3, 2024, and June 3, 2023, between 5$ and 85$. To achieve a profitable cross-exchange arbitrage trade between a CEX and an Ethereum-based DEX, it requires a high limit order volume x at a given spread level ρ to guarantee a profitable trade. Note, if x is low, it might result in unprofitable trade due to the fixed transaction cost $F_{\text{transaction}}$. Another problem can occur if the arbitrage trader quotes a high limit order volume x at a spread ρ that exceeds trading costs, as it cannot be guaranteed that the limit order will be completely filled. A fractional fulfillment of the limit order forces the arbitrage trader to trade-off between an unprofitable arbitrage trade and an unbalanced arbitrage portfolio. To circumvent the issue, one strategy is to include a CEX that uses a market order to guarantee a balanced arbitrage trade. However, a market order with a large volume at the CEX comes at the cost of an increased slippage at the CEX s_{CEX}. For blockchains that have a high transaction fee, arbitrage trading between two DEXs could be an alternative, as the expected slippage and order fulfillment can be predetermined more accurately. Concluding, the design of the arbitrage algorithm is heavily dependent on the cost function of the arbitrage environment. Expanding research into arbitrage trading strategies under different cost functions can provide a deeper understanding of practical arbitrage strategies.

2.3. Mempool calculations and transaction sequences

Trading at CEXs heavily depends on execution speed. The execution of trades on the blockchain is not only contingent on execution speed but also on the blockchain consensus mechanism and its corresponding transaction sequence. Borri and Shakhnov (2022) state that investors are exposed to execution risks that depend on time and the state of the network. The following Section shows that uncertainty about the spot price at the DEX remains if the transaction sequence of the blockchain is not known. Further, to receive accurate prices at the DEX $P_{\text{DEX}x_1/x_2}$, traders should not rely on the price data received from the smart contract. Instead, they should incorporate mempool transactions to determine the live price at the DEX. Transactions that are sent but not yet mined are located in the mempool.[10] Once a new block is found, the smart contract reserves are updated. The time delay and corresponding outdated smart contract reserves impose the risk of a high price deviation Δp_{DEX} at the DEX. The longer the average expected block time, the longer it takes to update the spot price which can lead to a higher risk of a spot price deviation. To avoid a time delay until the block is found, every transaction in the mempool requires to be analyzed if it interacts with the trading pair of interest. If a transaction has an influence on the price of the trading pair $P_{\text{DEX}x_1/x_2}$, it requires the arbitrage trader to compute the actual price $\hat{P}_{\text{DEX}x_1/x_2}$ at the DEX. Recall the conservation function from Section 2.1 and assume a mempool containing two raw transactions, a_0 at t_0 and a_1 at t_1. For the first trade a_0, the pool receives the amount x_1, and the trader receives the amount x_2. The new reserve level $\tilde{r}_{1,a_0}$ in the pool is calculated as (10) $\tilde{r}_{1,a_0}=r_1+x_{1,a_0}$ Then, the new level of reserve $\tilde{r}_{2,a_0}$ in the pool is determined from the conservation function as: (11) $K=\tilde{r}_{1,a_0}\cdot\tilde{r}_{2,a_0}=\tilde{r}_{2,a_0}=K/r_{1,a_0}=r_1\cdot r_2/r_1+x_{1,a_0}$ and the trader receives the output $x_2=\tilde{r}_{2,a_0}-r_2=r_1\cdot r_2\cdot(r_1-x_{1,a_0})/r_1+x_{1,a_0}$. The actual price after the trade a_0 is: (12) $\hat{P}_{\text{DEX}1/2,a_0}=\tilde{r}_{1,a_0}/r_{2,a_0}=(r_1+x_{1,a_0})/r_1\cdot r_2/r_1+x_{1,a_0}=(r_1+x_{1,a_0})^2/r_1\cdot r_2$ For the second trade a_1, the pool receives the amount x_2, and the trader receives the amount x_1. The new reserve level $\tilde{r}_{2,a_1}$ in the pool after the second trade is:
(13) $\tilde{r}_{2,a_1}=\tilde{r}_{2,a_0}+x_{2,a_1}=r_1\cdot r_2/r_1+x_{1,a_0}+x_{2,a_1}$ with the new level of reserve $\tilde{r}_{(1,a_1)}$ in the pool as:
(14) $K=\tilde{r}_{1,a_1}\cdot\tilde{r}_{2,a_1}=\tilde{r}_{1,a_1}=K/r_{2,a_1}=r_1\cdot r_2/r_1\cdot r_2/r_1+x_{1,a_0}+x_{(2,a_1)}$ and the trader receives the output $x_1=\tilde{r}_{1,a_1}-\tilde{r}_{1,a_0}=r_1\cdot r_2/r_1\cdot r_2/r_1+x_{1,a_0}+x_{2,a_1}-(r_1+x_{1,a_0})$. The actual price after the second trade a_1 is:

(15) $\hat{P}_{\text{DEX}1/2,a_1}=\tilde{r}_{1,a_1}/r_{2,a_1}=r_1\cdot r_2/r_1\cdot r_2/r_1+x_{1,a_0}+x_{2,a_1}/r_1\cdot r_2/r_1+x_{1,a_0}+x_{2,a_1}=r_1\cdot r_2(r_1\cdot r_2/r_1+x_{1,a_0}+x_{2,a_1})^2$ Steps from Eq. (10) to Eq. (15) show how to compute the actual price $\hat{P}_{\text{DEX}1/2,a_1}$ at the DEX before a new block is found based on fair ordering.[11] The arbitrage traders compute the actual price and adjust their order to keep their desired spread. Uncertainty remains if the sequence of trade execution is unknown.
Assume the order of these two raw transactions changes, so a_1 happens at t_0 and a_0 at t_1. This is more likely to happen if miners sort transactions based on their transaction fees, i.e., fee ordering, rather than fair ordering (timestamp-based). The order a_1 is sent at t_1 but it is executed first if $F_{\text{transaction}}$ of a_1 is larger than $F_{\text{transaction}}$ of a_0 that is sent at t_0. Then, for the first trade a_1, the pool receives the amount of x_2, and the trader receives the amount of x_1. The new level of reserve $\tilde{r}_{2,a_1}$ in the pool after the first trade is calculated as follows: (16) $\tilde{r}_{2,a_1}=r_2+x_{2,a_1}$ Then, the new level of reserve $\tilde{r}_{1,a_1}$ in the pool is: (17) $r_{1,a_1}=K/r_{2,a_1}=r_1\cdot r_2/r_2+x_{2,a_1}$ and the trader receives the output $x_1=\tilde{r}_{1,a_1}-r_1=r_1\cdot r_2\cdot(r_2-x_{2,a_1})/r_2+x_{2,a_1}$. The actual price after the first trade a_1 is: (18) $\hat{P}_{\text{DEX}1/2,a_1}=\tilde{r}_{1,a_1}/r_{2,a_1}=r_1\cdot r_2/r_2+x_{2,a_1}/r_2+x_{2,a_1}=r_1\cdot r_2(r_2+x_{2,a_1})^2$ For the second trade a_0, the pool receives the amount of x_1, and the trader receives the amount of x_2. The new level of reserve $\tilde{r}_{1,a_0}$ in the pool is: (19) $\tilde{r}_{1,a_0}=\tilde{r}_{1,a_1}+x_{1,a_0}=r_1\cdot r_2/r_2+x_{2,a_1}+x_{1,a_0}$ with the new level of reserve $\tilde{r}_{2,a_0}$ in the pool after the second trade as: (20) $\tilde{r}_{2,a_0}=K/r_{1,a_0}=r_1\cdot r_2/r_1\cdot r_2/r_2+x_{2,a_1}+x_{1,a_0}$ and the trader receives the output

$x_2 \tilde{=} r_{2,a_0} \tilde{-} r_{2,a_1} = r_1 \cdot r_2 r_1 \cdot r_2 r_2 + x_{2,a_1} + x_{1,a_0} - (r_2 + x_{2,a_1})$. The actual price after the second trade a_0 at $t=1$ is:

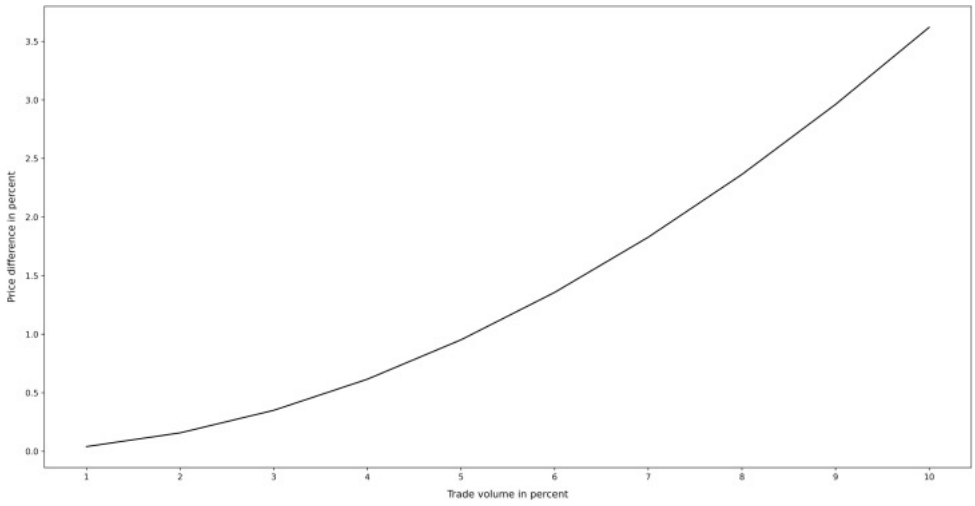
(21) $\hat{P}_{DEX1/2,a_0} \tilde{=} r_{1,a_0} / r_{2,a_0} = r_1 \cdot r_2 r_2 + x_{2,a_1} + x_{1,a_0} r_1 \cdot r_2 r_1 \cdot r_2 r_2 + x_{2,a_1} + x_{1,a_0} = (r_1 \cdot r_2 r_2 + x_{2,a_1} + x_{1,a_0})^2 r_1 \cdot r_2$

Steps from Eq. (16) to Eq. (21) show how to compute the actual price $\hat{P}_{DEX1/2,a_1}$ at the DEX before a new block is found based on fee ordering. Table 1 highlights the nuanced differences in spot prices based on the order sequence of the two trades, emphasizing how structural differences in the formulas affect price sensitivity and stability. In fair ordering, the numerator of the spot price represents the product of the initial reserves, which is a constant, while the denominator depends on the levels of x_{1,a_0} and x_{2,a_1}. This implies that any increase in x_{2,a_1} will decrease the denominator, thereby reducing the spot price. Moreover, the squared denominator amplifies this effect, making the spot price more sensitive to adjustments.
In contrast, under fee ordering, the numerator of the spot price depends on the levels of x_{2,a_1} and x_{1,a_0}; thus, any increase in x_{1,a_0} raises the numerator, increasing the spot price. Note that any increase in transaction fees can change the order sequence in fee ordering. For fair ordering, a change in order sequence is less likely. However, Baum et al. (2022) argue that even under fair ordering, front-running attacks remain a threat, as an attacker can rush transactions within the network.[12] Consequently, traders can never be fully certain of the final transaction sequence under either fair or fee ordering.

Table 1. Comparison of fair and fee ordering effects on spot prices.

Empty Cell	**First trade**	**Second trade**
Fair ordering:		
$a_0, t_0 \Rightarrow a_1, t_1$	$\hat{P}_{DEXx_1/x_{2,a_0}} = (r_1 + x_{1,a_0})^2 r_1 \cdot r_2$	$\hat{P}_{DEXx_1/x_{2,a_1}} = r_1 \cdot r_2 (r_1 \cdot r_2 r_1 + x_{1,a_0} + x_{2,a_1})^2$
Fee ordering:		
$a_1, t_1 \Rightarrow a_0, t_0$	$\hat{P}_{DEXx_1/x_{2,a_1}} = r_1 \cdot r_2 (r_2 + x_{2,a_1})^2$	$\hat{P}_{DEXx_1/x_{2,a_0}} = (r_1 \cdot r_2 r_2 + x_{2,a_1} + x_{1,a_0})^2 r_1 \cdot r_2$

Fig. 1 depicts the simulation of spot price deviation conditional on trade sequences. The following simulation assumes a trade volume for a_0 as x_1/r_1 and for a_1 as x_2/r_2. The trades are executed in the following order $[a_0, a_1]$ and $[a_1, a_0]$. The initial amount for reserve r_1 and reserve r_2 are 750,000 USDT and 10 BTC. We calculate the spot price after $[a_0, a_1]$ as $\hat{P}_{DEXx_1/x_{2,a_1}}$ and after $[a_1, a_0]$ as $\hat{P}_{DEXx_1/x_{2,a_0}}$, and the spot price change as $\Delta \hat{P}^{DEX} = \hat{P}_{DEXx_1/x_{2,a_1}} / \hat{P}_{DEXx_1/x_{2,a_0}}$. We repeat these calculations for varying trading volume where $x_1/r_1 = x_2/r_2$ at t_0 takes values in the interval $[0.01, 0.1]$, increasing the trading volume by 1 percent in each step.[13]
The y-axis depicts the spot price change in basis points for different volume levels. The results show that for a small trading volume, the order sequence has a minor influence. However, with rising trading volume, the relevance of the order sequence gains importance. For example, with highly volatile assets traded using a constant product AMM, a difference in the transaction sequence can result in significant spot price differences. Miners can exploit the transaction sequence to enhance their exposure position without engaging in any trading activity.



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Fig. 1. Spot price change.

To sum up, successful high-frequency trading at the CEX is primarily dependent on a good network connection. In contrast, trading at the DEX also depends on the blockchain's consensus mechanism, which determines the transaction sequence. This sequence can lead to front-running attacks, potentially resulting in unprofitable trades. Baum et al. (2022) propose fair ordering as a transaction sequencing method to reduce front-running attacks; however, as they state, even fair ordering cannot completely mitigate front-running. An attacker who identifies a trade in the mempool that significantly moves the DEX price by $s_{DEX}(x)$ can front-run the trader with a trade x_1 by paying an increased transaction fee $F_{transaction}(x_1)$. Additionally, the attacker can "sandwich" the trader by closing their position with a second trade x_2 to recover the initial asset x_1.

If the attacker increases their exposure position Δx_1 and the monetary value of $\Delta x_1 > F_{\text{transaction}}(x_1) + F_{\text{transaction}}(x_2)$, the sandwich attack is successful. To avoid becoming a victim of sandwich attacks, arbitrage traders can choose a trading volume such that $s_{\text{DEX}}(x)$ remains low, which reduces economic opportunities for attackers. Daian et al. (2019) describe how front-running and price gas auctions between multiple attackers have led to increased transaction fees on the Ethereum network. Solana recently introduced priority fees, which might pave the way for price gas auctions on Solana and lead to increased transaction costs in the future.[14]

3. Arbitrage trading bot

3.1. Setup

To set up an efficient arbitrage bot, a developer requires a good connection to the network. Garvey and Wu (2010) conduct an analysis of 2000 stock traders and find that being geographically closer to the exchange enhances execution speed and reduces execution costs. Additionally, they observe that traders situated nearer to the exchange's computers are more likely to engage in time-sensitive strategies. Unlike CEXs, DEXs run through blockchain validators that are geographically dispersed. Therefore, it is challenging to optimize the location to reduce execution speed. Beside the reduced latency and continuous network connection, a server-hosted blockchain node provides additional advantages over a local node. The server infrastructure offers a high level of security, significantly reducing the risk of cyber-attacks. For example, the Bitvise SSH client management system provides a secure means of connecting a local computer to a remote server.[15]

A server-hosted node offers a distinct economic advantage by eliminating the initial costs associated with purchasing hardware for setting up a node. It provides a flexible solution to scalability challenges. As blockchain software evolves, server-based solutions are advantageous in ensuring the infrastructure can adapt to growing resource demands. Summing up, a server infrastructure enhances the uptime, security, flexibility, and efficiency of the arbitrage system.

In addition to these economic and scalability benefits, maintaining a stable and up-to-date connection with the blockchain network is crucial for the arbitrage system's reliability. Since the DEX operates as a smart contract on the blockchain, the node automatically synchronizes with the network to retrieve the latest on-chain data. However, network disruptions or corrupt data can cause the node to lose connection to the blockchain network, leading to outdated price information from the DEX and preventing the execution of trades. In such cases, the trading bot should pause operations until the node is reconnected to the network. Restarting the node can often resolve the issue. In more severe situations, the current block data may need to be deleted, requiring a download of a network snapshot that contains the most recent data. It is crucial to obtain this snapshot from an official source to prevent the risk of malicious malware.

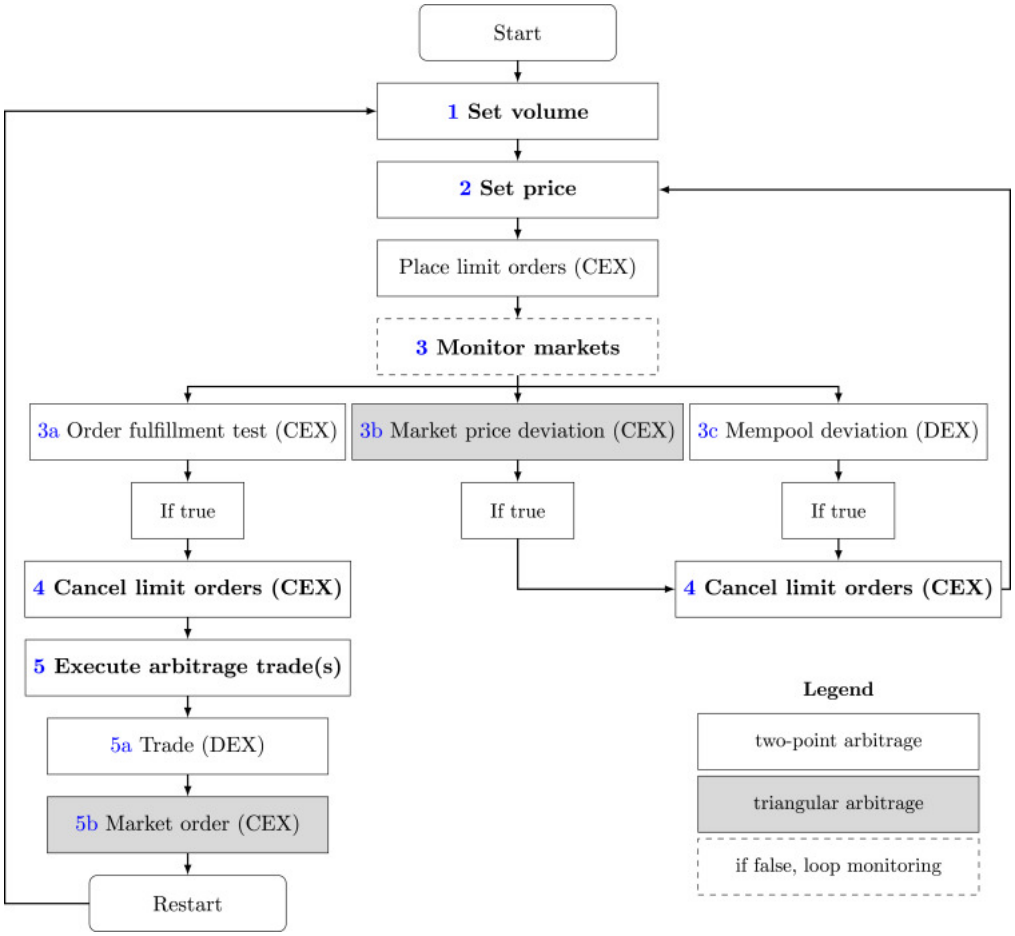
In contrast to the node that is hosted on the server to interact with the DEX, KuCoin operates as a CEX, which is privately managed and acts as a custodian for traders. Data is provided through RESTful API, enabling communication between the server and client. To authenticate the server with the client, users must generate an API key, API secret, and passphrase. To further enhance security, KuCoin allows the configuration of IPv6 addresses to restrict data access to specified addresses. In this setup, we configure the server's IPv6 address as the only permitted address for API connections, adding an extra layer of security by preventing unauthorized withdrawals, even if API credentials are compromised.

Additionally, we use Crontab, which enhances the arbitrage system in several ways. First, for reporting purposes, a separate Python script is deployed on the server to document the trading bot's overall exposure position, with updates sent to the private Telegram group every minute. Second, KuCoin reduces trading fees by 20 percent for users holding KuCoin tokens (KCS). When the KCS function is enabled, trading fees are deducted from the KCS balance. Another Python script is scheduled to monitor the KCS balance and automatically purchase additional KCS if it falls below a preset threshold. This ensures a minimized KCS exposure position while maintaining the 20 percent fee reduction. Third, for every triangular arbitrage trade, the system tracks the DEX order ID and the limit or market order IDs from the CEX. A further Python script retrieves trade data from each exchange using these IDs to calculate the realized profit.

3.2. Algorithm

In this Subsection, we outline the order placement and execution strategy for two-point and triangular arbitrage between a DEX and a CEX. We outline the design of the trading bots, detailing the conditions for canceling a quote, managing limit order fulfillment, and how the bot's algorithm can minimize slippage costs. Additionally, we discuss the algorithmic distinctions between two-point and triangular arbitrage strategies, along with the calculations performed at each step. In Section 4, we present the practical outcomes of the trading bots. The arbitrage system on the SEI blockchain executes two-point arbitrage, while the system on the DeFiChain employs triangular arbitrage. In the following, the algorithms are explained in relation to the corresponding trading pairs at the DeFiChain but are adaptable to other trading pairs across cryptocurrency markets.

Fig. 2 depicts the design of the arbitrage bot in six sequential steps, where the two-point arbitrage strategy operations are marked in white boxes, and the triangular arbitrage bot's additional computational tasks are marked in gray boxes. The reason for the triangular arbitrage strategy is the unavailability of equal trading pairs at both exchanges. For our trading bot, we aim to arbitrage the DFI-SOL pool at the DEX. However, the DEX is the only exchange that lists that specific trading pair. Therefore, we mimic the DFI-SOL trading pair with two trading pairs at the CEX, i.e., DFI-USDT and SOL-USDT. For example, a direct trade of DFI sold to SOL at the DEX is balanced by selling SOL to USDT to use USDT to purchase an equivalent amount of DFI at the CEX. Any bot design that necessitates executing a market order within a limit order book system is inherently risky due to unpredictable slippage costs. If the liquidity of a market is low, the slippage costs increase.



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Fig. 2. Bot design.

As a preparatory step, it is essential to assess the liquidity depth in the DFI-USDT and SOL-USDT markets at the CEX. Brauneis et al. (2021) describe and compare high-frequency, easy-to-compute liquidity measures to develop liquidity proxies. A separate Python script retrieves order book data on a minute-by-minute basis to calculate the expected slippage for various trade volumes, as described in Section 2. Unlike the SOL-USDT market, which has sufficient liquidity depth, we identify that the DFI-USDT market exhibits low liquidity, which makes it costly to work with a market order. Therefore, we use a market order in the SOL-USDT market and a limit order in the DFI-USDT market.
In the following, we describe the bot design, referring to Fig. 2 in every detail. In order to make the design of the arbitrage bot as understandable as possible, we include an example for the triangular arbitrage strategy. For this, assume the arbitrage portfolio holds positions as shown in Table 2. Additionally, we set the maximum trading volume ($\max_{\text{vol}}$) at the level of 1000 USDT and the minimum trading volume ($\min_{\text{vol}}$) at the level of 10 USDT. The desired bid and ask spread (ρ) is at the level of 120 basis points. Additionally, fees at the CEX and at the DEX ($f_{\text{CEX}_{\text{limit}}}$ and f_{DEX}) amount to 20 basis points, fees for the market order at the CEX ($f_{\text{CEX}_{\text{market}}}$) amounts to 10 basis points. Slippage at the CEX and DEX (s_{CEX} and s_{DEX}) amount to 5 and 10 basis points respectively.

Table 2. Arbitrage portfolio and prices.

Empty Cell	**DEX**	**CEX**
Exposure:		
SOL	30	70
DFI	10,000	3000
USDT		1,500
Prices:		
DFI-SOL	0.004	
SOL-USDT		50
DFI-USDT		0.2

Note: This table shows exposure positions and spot prices at the DEX and CEX. These current spot prices and exposure position are the basis for the example calculation through out the Section 3.

- 1.
Set volume: To avoid slippage costs, the bot algorithm starts with two limit orders in the DFI-USDT market at the CEX ($P_{\text{CEXDFI-USDT}}$) that wait to be executed. A bid ($P_{\text{CEX}}^{\text{bid}}{}_{\text{DFI-USDT}}$) and ask ($P_{\text{CEX}}^{\text{ask}}{}_{\text{DFI-USDT}}$) limit orders are placed around the spot price to enable a limit order fulfillment if the market moves in either direction. Placing

two limit orders requires determining the appropriate limit order prices ($P_{\text{CEX}}^{\text{ask}}{}_{\text{DFI-USDT}}$ and $P_{\text{CEX}}^{\text{bid}}{}_{\text{DFI-USDT}}$) and volumes ($\text{bid}_{\text{vol}}$ and ask_{vol}). The exposure positions on both exchanges decide upon the size of the trade volume. For the bid order, the trader aims to buy DFI liquidity (L_{CEXDFI}) at the CEX with USDT funds (L_{CEXUSDT}). Balancing the two-point arbitrage trade requires an equivalent USDT value in DFI ($L_{\text{DEXDFI}} \cdot P_{\text{CEXDFI-USDT}}$) to be sold at the DEX. Therefore, the maximum possible trade volume is determined by the minimum value of different exposure positions and a maximum volume cap designed to limit slippage costs. The volumes for two-point arbitrage are calculated as follows:

$$\text{bid}_{\text{vol}} = \min(\text{CEX}\ \ L_{\text{USDT}}, L_{\text{DEXDFI}} \cdot P_{\text{CEXDFI-USDT}}, \max_{\text{vol}}) \geq \min_{\text{vol}} \quad (22)$$

$$\text{ask}_{\text{vol}} = \min(\text{DEX}\ \ L_{\text{USDT}}, L_{\text{CEXDFI}} \cdot P_{\text{CEX}} \quad (23)$$

The volumes for triangular arbitrage are calculated as follows:

$$\text{bid}_{\text{vol}} = \min(\text{CEX}\ \ L_{\text{USDT}}, L_{\text{DEXDFI}} \cdot P_{\text{CEXDFI-USDT}}, L_{\text{CEXSOL}} \cdot P_{\text{CEXSOL-USDT}}, \max_{\text{vol}}) \geq \min_{\text{vol}} \quad (24)$$

$$\text{ask}_{\text{vol}} = \min(\text{C} \quad (25)$$

For the triangular bid arbitrage trade, an additional constraint is the SOL position at the CEX ($L_{\text{CEXSOL}} \cdot P_{\text{CEXSOL-USDT}}$). At the CEX, SOL liquidity (L_{CEXSOL}) is sold and DFI liquidity (L_{CEXDFI}) is purchased. Meanwhile, at the DEX, DFI L_{DEXDFI} is directly sold to purchase SOL (L_{DEXSOL}). Based on the arbitrage portfolio shown in Table 2, the example calculation illustrates that the bid and ask volumes are determined by different underlying constraints when computing the quote volume. In this scenario, the arbitrage bot operates with a bid volume of 1000 USDT and an ask volume of 600 USDT. Note, despite the available liquidity at either exchange, the maximum and minimum trading volume, i.e., $\max_{\text{vol}}$ and $\min_{\text{vol}}$, for either limit order cannot be infinitely large or close to zero. If either volume falls below 10 USDT, i.e., $\min_{\text{vol}}$, no quote is placed. Recall from Section 2.2, an increased trading volume increases the slippage cost and the risk of being sandwiched at the DEX. Therefore, maximum trading volume controls for excessive slippage costs. As KuCoin defines a minimum trade volume for their markets, $\min_{\text{vol}}$ ensures that no order is placed if the calculated trading volume falls below this threshold.

- 2.
  Set price: The second parameter for placing a limit order is calculating the limit order prices. For two-point arbitrage, to compute the limit order price, the bot retrieves the price for DFI ($P_{\text{DEXDFI-USDT}}$) at the DEX and adds a configurable spread level ρ. For triangular arbitrage, it requires the price of Solana from the CEX ($P_{\text{CEXSOL-USDT}}$) to express the DFI ($P_{\text{DEXDFI-SOL}}$) in USDT. The prices for two-point arbitrage are calculated as follows:

  $$P_{\text{CEX}}^{\text{bid}}{}_{\text{DFI-USDT}} = P_{\text{DEXDFI-USDT}} \cdot (1-\rho) \quad (26)$$

  $$P_{\text{CEX}}^{\text{ask}}{}_{\text{DFI-USDT}} = P_{\text{DEXDFI-USDT}} \cdot (1+\rho) \quad (27)$$

  The prices for triangular arbitrage are calculated as follows:

  $$P_{\text{CEX}}^{\text{bid}}{}_{\text{DFI-USDT}} = P_{\text{DEXDFI-SOL}} \cdot P_{\text{CEXSOL-USDT}} \cdot (1-\rho) \quad (28)$$

  $$P_{\text{CEX}}^{\text{ask}}{}_{\text{DFI-USDT}} = P_{\text{DEXDFI-SOL}} \cdot P_{\text{CEXSOL-USDT}} \cdot (1+\rho) \quad (29)$$

  $$P_{\text{CEX}}^{\text{bid}}{}_{\text{DFI-USDT}} = 0.004 \cdot 50 \cdot (1-0.012) = 0.1976 \text{USDT} P$$

  Given a specified price and volume, bid and ask limit orders can be placed in the limit order book. In the given example, the bid limit order is placed with a volume of 1000 USDT at a price of 0.1976 USDT. Similarly, the ask limit order is placed with a volume of 600 USDT at a price of 0.2024 USDT.
- 3.
  Monitor markets: The bot starts the price monitoring process and consists of the following processes:
  - (a)
    Order fulfillment test (CEX)
  - (b)
    Market price deviation (CEX)
  - (c)
    Mempool deviation (DEX)

  For two-point arbitrage, two parallel processes start with testing for a limit order fulfillment 3a and price deviations at the DEX that arise through orders in the mempool 3c. Any significant price change at the DEX must be handled to maintain the demanded spread level. To avoid the time delay until a new block is found, the bot predetermines the current price at the DEX. Refer to Section 2.3, which explains how to calculate the effect of trades on the spot price with the help of the conservation function. The bot monitors the mempool and tests for interactions with the DFI-USDT trading pool. For triangular arbitrage, three processes start 3a, 3b, and 3c. The triangular arbitrage bot requires an additional test 3b for the price change in SOL ($P_{\text{CEXSOL-USDT}}$) at the CEX. This is because the DFI price at the DEX depends on price changes in SOL. For the bot configuration, we define the price deviation threshold of 5 basis points. An increased threshold results in fewer order cancellations and extends the duration that orders remain active within the order book system. However, the trade-off is that the realized profit may exhibit greater variability due to the increased tolerance for price deviations, which can lead to less profitable outcomes. The prices reset for two-point arbitrage occurs when:

  $$3\text{cabs}(\Delta P_{\text{DEXDFI-USDT}}) > 5\text{basis points} \quad (30)$$

  The prices reset for triangular arbitrage occurs when:

  $$3\text{cabs}(\Delta P_{\text{DEXDFI-SOL}}) > 5\text{basis points or } 3\text{babs}(\Delta P_{\text{CEXSOL-USDT}}) > 5\text{basis points} \quad (31)$$

- 4.
  Cancel limit orders (CEX): If either condition 3b or 3c is met, old limit orders are canceled, and new limit orders are placed to ensure the initial spread level is maintained. The bot can continue using the same volume data, as there has been no change in the exposure position. The bot exits the looped processes if the limit order is filled 3a and the trade is rebalanced. Frequently, only a fraction of the limit order is filled. If the rest of the limit order is not canceled, either limit order might have additional fillings during the trade handling. To avoid a second filling and guarantee a balanced exposure position, both limit orders are canceled, and the filled limit order is handled. Please note that at the CEX, the trade volume for the SOL-USDT and DFI-USDT markets is measured in terms of SOL and DFI, respectively. In contrast, the DEX specifies the trade volume based on the input currency.
- 5.
  Execute arbitrage trade(s): The arbitrage execution starts with the fulfillment of the limit order at the CEX. This first trade is identical for both the two-point and the triangular arbitrage bot. For bid fulfillment, the limit order buys DFI with USDT ($\text{USDT}_{\text{CEXsell}_{\text{vol}}}$), while for ask fulfillment, it sells DFI, receiving USDT ($\text{USDT}_{\text{CEXbuy}_{\text{vol}}}$):

$$(32)\text{USDT}_{\text{CEXsell(buy)}_{\text{vol}}}=\text{DFI}_{\text{CEXbid(ask)}_{\text{fulfillment}}}\cdot P_{\text{CEX}}^{\text{bid(ask)}}{}_{\text{DFI-USDT}}\quad(33)\text{Fee}_{\text{CEX}_{\text{limit}}}=\text{USDT}_{\text{CEXsell(buy)}_{\text{vol}}}\cdot f$$

After limit order is filled, the bot executes following trades:

- (a) Trade (DEX)
- (b) Market order (CEX)

The two-point arbitrage bot executes only 5a after the limit order at the CEX has been filled. For bid order fulfillment, the two-point arbitrage bot executes a trade at the DEX, selling DFI to refill the USDT spent in the limit order:

$$(34)\text{USDT}_{\text{DEXbuy}_{\text{vol}}}=\text{DFI}_{\text{DEXbid}_{\text{fulfillment}}}\cdot P_{\text{DEXDFI-USDT}}\cdot(1-s_{\text{DEX}})\underbrace{\qquad}_{\text{Slippage}_{\text{DEX}}}\cdot(1-f_{\text{DEX}})\underbrace{\qquad}$$

ask order fulfillment, the two-point arbitrage bot executes a trade at the DEX, buying DFI with USDT received from the limit order:

$$(35)\text{USDT}_{\text{DEXsell}_{\text{vol}}}=\text{DFI}_{\text{CEXask}_{\text{fulfillment}}}\cdot P_{\text{DEXDFI-USDT}}(1-s_{\text{DEX}})\underbrace{\qquad}_{\text{Slippage}_{\text{DEX}}}\cdot(1-f_{\text{DEX}})\underbrace{\qquad}$$

The triangular arbitrage bot executes 5a and 5b after the limit order is filled at the CEX. For bid order fulfillment, the triangular arbitrage bot first executes a trade at the DEX, selling DFI and receiving SOL:

$$(36)\text{SOL}_{\text{DEXbuy}_{\text{vol}}}=\text{DFI}_{\text{CEXbid}_{\text{fulfillment}}}\cdot P_{\text{DEXSOL-USDT}}\cdot(1-s_{\text{DEX}})\underbrace{\qquad}_{\text{Slippage}_{\text{DEX}}}\cdot(1-f_{\text{DEX}})\underbrace{\qquad}$$

final trade is the execution of a market order at the CEX, selling SOL to refill the USDT spent in the limit order:

$$(37)\text{USDT}_{\text{CEXbuy}_{\text{vol}}}=\text{SOL}_{\text{CEXsell}_{\text{vol}}}\cdot P_{\text{CEXSOL-USDT}}\quad(38)\text{USDT}_{\text{CEXreceived}_{\text{vol}}}=\text{USDT}_{\text{CEXbuy}_{\text{vol}}}\cdot(1-s_{\text{CEX}})\underbrace{\qquad}$$

For ask order fulfillment, the triangular arbitrage bot first executes a trade at the DEX, selling SOL and receiving DFI:

$$(40)\text{SOL}_{\text{DEXsell}_{\text{vol}}}=\text{DFI}_{\text{CEXask}_{\text{fulfillment}}}\cdot P_{\text{DEXSOL-DFI}}(1-s_{\text{DEX}})\underbrace{\qquad}_{\text{Slippage}_{\text{DEX}}}\cdot(1-f_{\text{DEX}})\underbrace{\qquad}$$

final trade is the execution of a market order at the CEX, buying SOL with the USDT received from the limit order:

$$(41)\text{USDT}_{\text{CEXsell}_{\text{vol}}}=\text{SOL}_{\text{CEXbuy}_{\text{vol}}}\cdot P_{\text{CEXSOL-USDT}}(1-s_{\text{CEX}})\underbrace{\qquad}_{\text{Slippage}_{\text{CEX}}}\quad(42)\text{Slippage}_{\text{CEX}}=\text{USDT}_{\text{CEXs}}$$

For the triangular arbitrage strategy, the maximum trading volume requires limits for two reasons. First, as with two-point arbitrage, the slippage at the DEX should be limited. Second, to avoid high slippage cost for the market order at the CEX. The DEX deducts the trading fee from the trade output, unlike the CEX where the trading fee is subtracts from the balance. Following the trading activity, balances and prices at each exchange vary, demanding a recalculation of the quote variables. Given its new trading volume, the bot restarts the initial quotation and monitoring processes. Similarly, as with transactions in 3c, the bot automatically incorporates its own mempool transaction, adjusts the spot price, and operates with the updated DEX price accordingly. In case the bot runs out of liquidity for either coin at either exchange, a separate script manages the distribution of liquidity. The script is outsourced to avoid additional computations for the trading bot and increase the speed of the bot.

From the example above, the bid limit order with a volume of 1000 USDT at a price of 0.1976 aims to purchase 5060 DFI at the CEX. Assume that 4000 out of the 5060 DFI are executed through this limit order. Based on Eqs. (32), (33), the corresponding amount paid at the CEX is:

$$\text{USDT}_{\text{CEXsell}_{\text{vol}}}=4000\cdot0.1976=790.4\text{USDT}\quad f_{\text{CEX}_{\text{limit}}}=790.4\cdot0.002=1.58\text{USDT}$$

For this order execution, KuCoin charges a fee of 1.58 USDT, which corresponds to 20 basis points of the traded volume. Since a limit order executes at a fixed price, there is no slippage. In total, the bot spends 791.98 USDT for this trade.

After fulfilling the limit order, the bot sells the 4000 DFI at the DEX to rebalance its DFI position. As DEX trades are executed only when a block is mined, we need to predict the amount of SOL that will be received at the DEX. This allows the bot to complete the triangular arbitrage by selling the same amount of SOL at the CEX. The expected amount of SOL received at the DEX (is equivalent to the amount of SOL sold at the CEX), with respect to Eq. (36), is calculated as follows:

$$\text{E}(\text{SOL}_{\text{DEXbuy}_{\text{vol}}})=\text{SOL}_{\text{CEXsell}_{\text{vol}}}=4000\cdot0.004\cdot(1-0.001)\underbrace{\qquad}_{s\text{DEX}}\cdot(1-0.002)\underbrace{\qquad}_{f\text{DEX}}=15.9$$

trade output at the DEX is subject to a slippage of 10 basis points, which can be deterministically computed using Eq. (3). In this trade, the DEX trade incurs a slippage of 0.016 SOL (0.8 USDT) and trading fees of 0.032 SOL (1.6 USDT). Unlike at the CEX, where trading fees are externally handled and deducted from the account balance in USDT, trading fees at the DEX are subtracted directly from the trade output. To complete the triangular arbitrage, the bot must sell the received SOL at the CEX to refill its USDT balance. Based on Eqs. (37), (38) and (39), the bot sells 15.95 SOL using a market order at the CEX, yielding:

$$\text{USDT}_{\text{CEXbuy}_{\text{vol}}}=15.95\cdot50=797.6\text{USDT}\quad\text{USDT}_{\text{CEXrecieved}}=797.6\cdot(1-0.0005)\underbrace{\qquad}_{s\text{CEX}}=797.2\text{USDT}\quad f_{\text{CEXma}}$$

The bot experiences a slippage of 5 basis points at the CEX, which corresponds to 0.008 SOL (0.4 USDT). Additionally, the market order fee at the CEX amounts to 10 basis points of the trading volume, or 0.8 USDT. After accounting for slippage and fees, the bot receives 796.4 USDT from the market order. Thus, the total cost of executing the triangular arbitrage trade, as defined in Eq. (7), is:

$$c_{triangular}\text{trade}=\underbrace{20}_{f\text{CEX}_{\text{limit}}}+\underbrace{10}_{f\text{CEX}_{\text{market}}}+\underbrace{20}_{f\text{DEX}}+\underbrace{5}_{s\text{CEX}}+\underbrace{10}_{s\text{DEX}}=65\text{basis points}$$

Assume that during the execution of this triangular trade, there is no spot price change for the market order at the CEX ($\Delta p\text{CEX}=0$) and no spot price change for the trade at the DEX ($\Delta p\text{DEX}=0$). Further, assume there are no on-chain transaction costs ($F\text{transaction}=0$). The realized arbitrage spread after all trades amounts to:

$$\text{Net profit}=-791.98+796.4=4.4\text{USDT}\quad\text{Realized spread}=\frac{4.4}{797.6}=0.0055=55\text{basis points}$$

The limit order at the CEX costs the arbitrage bot 791.98 USDT, and after the market order trade at the CEX, the bot receives 796.4 USDT. Thus, the net profit from this triangular arbitrage trade is 4.4 USDT. The trading volume of the market order before slippage is 797.6 USDT, resulting in a realized spread of 55 basis points. The profit of this triangular trade, based on Eq. (8), amounts to:

$$\pi=(0.012-0.0065-0-0)\underbrace{\qquad\qquad}\quad55\text{basis points}\cdot797.6-0=4.4\text{USDT}$$

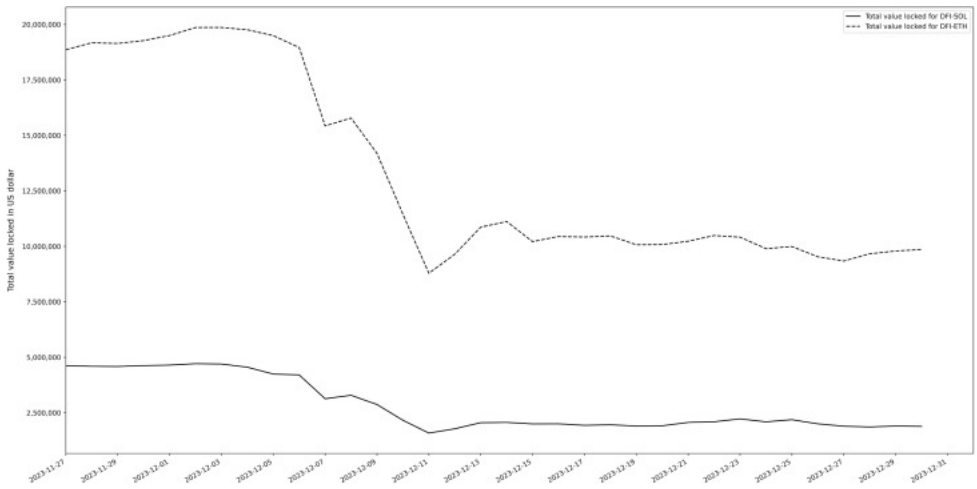
This example demonstrates that an arbitrage bot that accounts for slippage and trading fees can accurately estimate the realized spread. To perfectly balance the triangular arbitrage trade, the bot must deterministically compute the expected amount received at the DEX.

The results of our trading bot presented in Section 4 and Section 5 demonstrate the efficiency of the algorithm in executing profitable trades. Nevertheless, unprofitable outcomes for triangular arbitrage strategies can occur, as described in the following.

First, the quoted limit-order price at the CEX depends on two reference prices: (i) the price at the DEX (0.004 DFI-SOL) and (ii) the price at the CEX (50 SOL-USDT) that executes the market order. The price at the DEX is determined by the reserve

levels within the smart contract and the transaction sequence on the blockchain. If the blockchain node loses synchronization and operates with outdated reserve data or the transaction sequences is unknown, the limit-order price at the CEX (0.1976 DFI-USDT) becomes mispriced. Executing such a limit order may, therefore, result in an unprofitable trade.
Second, frontrunning attacks remain a threat to arbitrage traders during the execution of trade cycles. If the arbitrage trader becomes the victim of a frontrunning attack or if other traders execute their transactions slightly earlier the resulting DEX output decreases ($E(SOL_{DEXbuy_{vol}}) < 15.95$ SOL), thereby reducing the trader's expected profit.

Third, frontrunning liquidity provision attacks at the DEX can further diminish the expected trade output ($E(SOL_{DEXbuy_{vol}}) < 15.95$ SOL). The deterministic pricing model at the DEX allows the calculation of slippage ($s_{DEX} = 10$ basis points), which represents a cost factor for the arbitrage trader that increases as liquidity within the smart contract reserves decline. While the arbitrage trader awaits the execution of their order, an attacker may subsequently withdraw a substantial portion of the DEX liquidity and prioritize their transaction by paying a higher fee. The attacker then closes their position through a counter trade after the arbitrage transaction is executed. When the block is mined, the transaction sequence can unfold as follows: (1) liquidity is first withdrawn by the attacker; (2) the arbitrage transaction executes with increased slippage costs ($s_{DEX} > 10$ basis points); (3) the attacker closes their position with a trade, buying/selling an asset that got mispriced by the arbitrageurs who triggered a high spot price change through their high slippage. Consequently, the arbitrage trader faces increased slippage, higher trading costs, and potentially unprofitable results.



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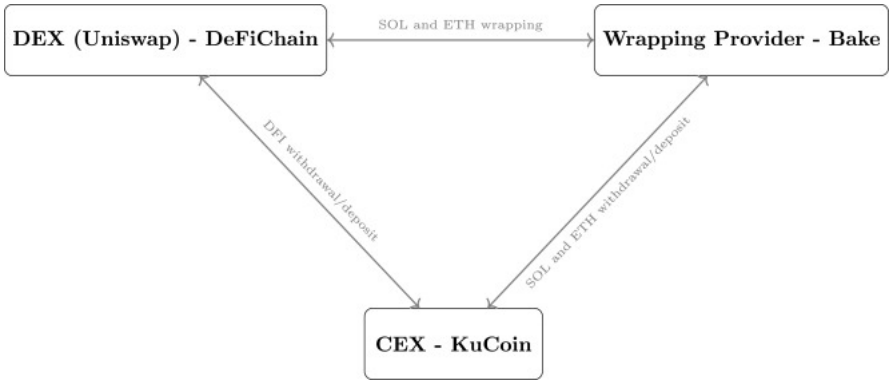
Fig. 3. Total value locked on DeFiChain.

Fourth, slippage and price deviation risks also exist at the CEX. Any price change ($\Delta P^{CEX} > 0$) for the market order can alter the executed price, and any liquidity withdrawal from the order book increases slippage costs ($s_{CEX} > 5$ basis points). Both effects can contribute to unprofitable arbitrage trades.
Fifth, for blockchains with highly volatile transaction fees, sudden spikes in network fees can quickly erode or eliminate profits for the arbitrage trader.

4. Empirical analysis

The identification of an appropriate arbitrage environment depends on several aspects. Specifically, the degree of market fragmentation is closely related to the prevalence of arbitrage opportunities. The more fragmented a capital system is, the more arbitrage opportunities it offers.
DEXs which are well embedded in a capital system through DEX aggregators typically exhibit fewer arbitrage opportunities. This is because aggregators automatically route traders to exchanges offering the highest trade outputs, namely, to those with spot prices that deviate significantly from market prices across all DEXs. This process automatically facilitates price adjustments on aggregator integrated exchanges. Therefore, these DEXs experience higher trading volumes, rapidly aligning their spot prices with market prices. On the contrary, DEXs which are not integrated into aggregators do not benefit from such automatic price corrections. Without these noise traders adjusting prices, deviations between spot and market prices persist, leading to more frequent arbitrage opportunities. The same mechanism as that of aggregators exists within individual DEXs. When a trader operates on a single exchange, smart order routing directs the trader to the pool offering the highest trade output on that DEX. As a result, noise traders are routed to pools with the largest price deviations, facilitating automatic price correction within the exchange.
Available liquidity is another key factor in selecting an arbitrage environment, as low liquidity leads to high slippage costs, which constraints the volume an arbitrage trader can execute. When market prices change, deeper liquidity in a DEX trading pair allows arbitrage traders to capitalize on opportunities more effectively by executing larger trades.
After selecting exchanges for the arbitrage system, it is important to measure the cross-exchange spread of the chosen trading pairs to assess the extent of available arbitrage opportunities. However, the spread alone cannot serve as a universal benchmark, as transaction costs vary across exchanges, wrapping providers, and blockchains. Therefore, the measured cross-exchange spreads must be evaluated relative to the costs outlined in Section 2.2. The focus lies on profit-making spreads that enable arbitrage opportunities. Notably, such spreads are typically larger for trading pairs not listed on CEXs and often require more complex algorithmic strategies, as detailed in Section 3.2.
We identify that the trading pairs DFI-ETH and DFI-SOL are exclusively available on the DeFiChain DEX. To replicate these pairs on a CEX, a triangular arbitrage strategy is required, involving the DFI-USDT and ETH-USDT pairs for the DFI-ETH trading pair, and the DFI-USDT and SOL-USDT pairs for the DFI-SOL trading pair. Both trading pairs at the DEX offer

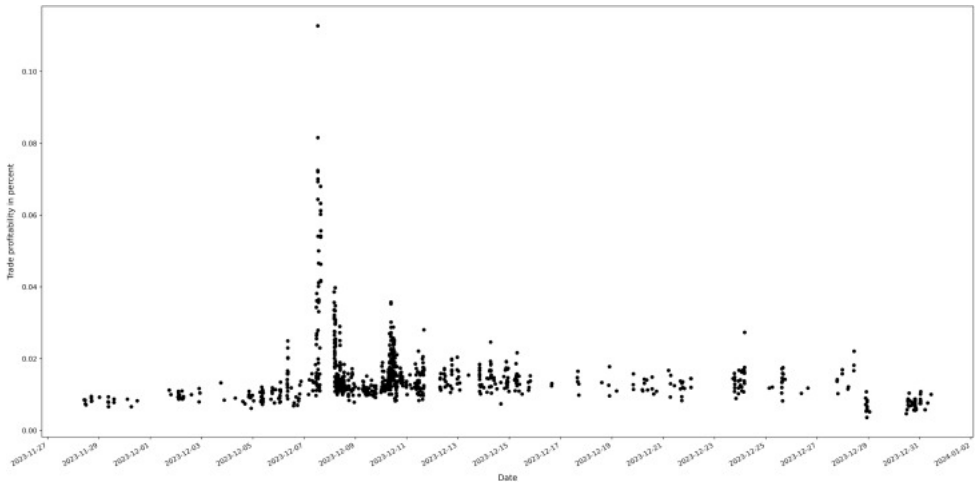
sufficient liquidity and regularly exhibit profit-making spreads. Fig. 3 displays the total value locked (TVL), in US dollars, for the DFI-ETH and DFI-SOL trading pairs during the operating period of the arbitrage system. The TVL amounts to $18.8 million for the DFI-ETH trading pair and $4.5 million for the DFI-SOL trading pair. The combination of sufficient liquidity and relatively low competition motivates the decision to deploy our initial arbitrage system on the DeFiChain DEX.
Fig. 4 depicts the liquidity flow for DFI, SOL, and wrapped SOL (wSOL) on the DeFiChain. Caldarelli (2021) explains that wrapped tokens help to overcome the absence of communication between blockchains. The DeFiChain has no direct connection with the Solana blockchain, which requires a wrapping provider to move assets between blockchains. If SOL liquidity runs low at the DEX, the funds are circulated through the wrapping provider (native SOL transaction sends the liquidity from KuCoin to Bake). Bake Ltd serves as a custodian for native crypto assets on the DeFiChain. It locks the SOL on the Solana blockchain and mints an equivalent amount of wSOL on the DeFiChain. The wSOL tokens are sent via a native DeFiChain transaction from Bake to the bot address to increase wSOL liquidity at the DEX. The approach to use a centralized company as a custodian for all crypto assets bears a high risk for the blockchain and questions the decentralization of the blockchain. To circulate the native DFI coin, a direct withdrawal from Kucoin to DeFiChain and vice versa helps to keep a balanced distribution of DFI funds.



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Fig. 4. Flow between DEX, wrapping provider, and CEX.

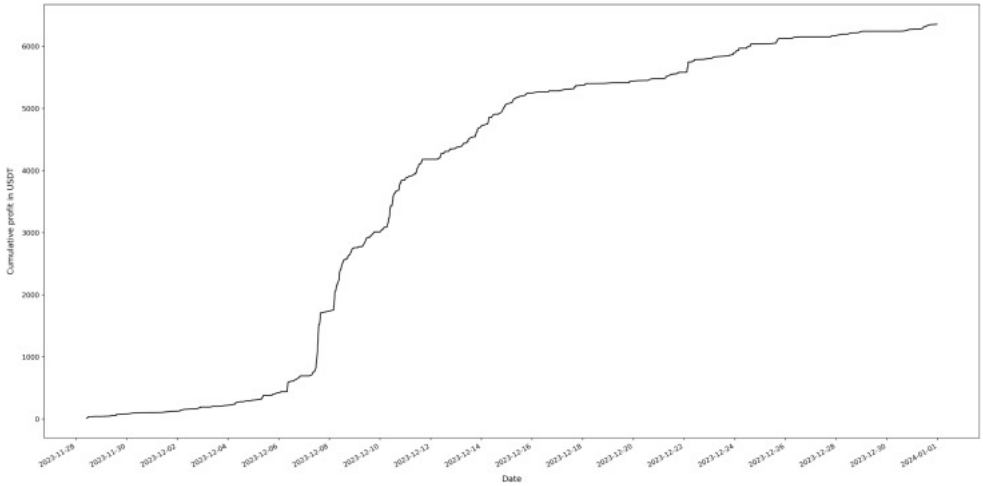
The trading period starts on November 27, 2023, and lasts until December 31, 2023. We deploy the arbitrage system on two different wallet addresses.[16] The fee rate for the limit order amounts to $f_{\text{CEX}_{\text{limit}}}=16$ basis points and for the market order $f_{\text{CEX}_{\text{market}}}=8$ basis points. The fee rate at the DEX is $f_{\text{DEX}}=20$ basis points.[17] The spread level is configured at $\rho=120$ basis points. The first triangular arbitrage system, dMFP, executes arbitrage trades for the following trading pairs: DFI-USDT, SOL-USDT, and DFI-SOL. The second triangular arbitrage system, dD6K, runs two parallel trading bots. The first bot operates on the DFI-USDT, SOL-USDT, and DFI-SOL pairs, while the second bot executes arbitrage trades for the trading pairs DFI-USDT, ETH-USDT, and DFI-ETH.
The initial exposure portfolio for the dMFP is 25,000 DFI, 2100 USDT, and 225 SOL. The maximum trade volume for each transaction is capped at $1500 to reduce slippage costs at both exchanges, i.e., $s_{\text{CEX}}+s_{\text{DEX}}$. Based upon the calculations described in Section 2.1, the expected slippage for a trade volume of $1500 at the CEX, $\text{E}(s_{\text{CEX}})$, ranged from 0 to 2 basis points and at the DEX, $\text{E}(s_{\text{DEX}})$, ranged from 0 to 1 basis points. The variation in slippage at the CEX is attributed to the fluctuating liquidity available, which changes frequently. In contrast, the variation at the DEX is primarily influenced by the TVL in the corresponding trading pair. Although liquidity deposits remain mostly stable, the TVL fluctuates with changes in the price of underlying assets. At the start and end of the trading period, the coin prices on November 27 were: SOL $59.18, DFI $0.246, ETH $2064.07, USDT $1, and on December 31: SOL $101.99, DFI $0.159, ETH $2294.34, USDT $1.
Fig. 5 illustrates the distribution of realized arbitrage profit in percent for the dMFP arbitrage system. The realized profit incorporates the trading costs C but excludes the cost C_{travel} that occurs whenever liquidity rotates between the exchanges. Each data point represents the realized profit for a triangular arbitrage trade π in percent. The dMFP targets an expected profit of 73–76 basis points (based on realized spread in Eq. (8)). Fig. 5 shows outliers up to 1070 basis points. In moments of high volatility, the quoted bid/ask order price is higher/lower than the best ask/bid price in the order book, and two limit orders automatically converge at the best ask/bid price, which allows for profits above 76 basis points.



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Fig. 5. Trade profitability dMFP.

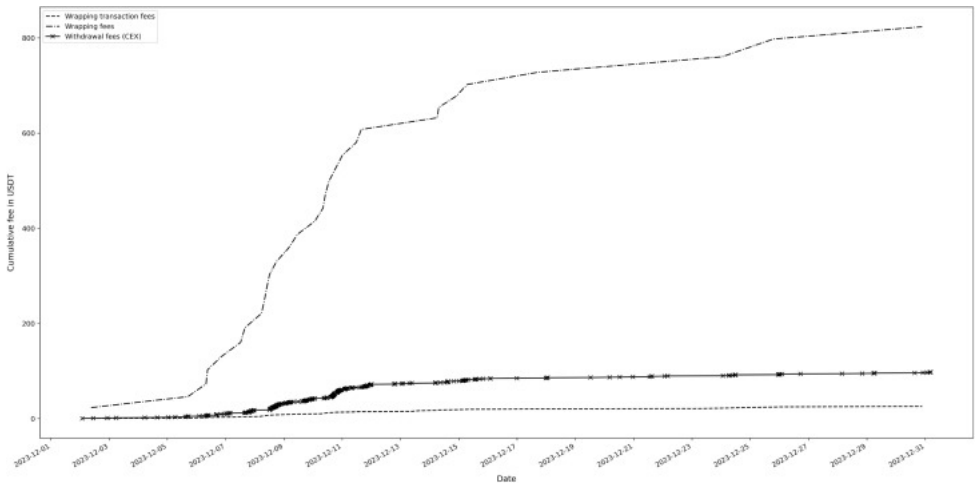
Fig. 6 shows the cumulative profit of the dMFP arbitrage system over time and depicts a continuous pattern of realized arbitrage opportunities. The total amount of profits is 6357.75 USDT. The most profitable trade yields a profit of 1070 basis points, amounting to $160.63 with a single arbitrage trade. For the dMFP, out of 1074 arbitrage trades, not a single trade results in a loss, which indicates a low execution risk. The average realized spread is 154 basis points and $5.88 profit per triangular arbitrage trade.



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Fig. 6. Cumulative profit dMFP.

To evaluate the overall performance of the bot, Fig. 7 shows the costs that occur through the rotation of liquidity. The travel cost C_{travel} incorporates the withdrawal cost $F_{\text{withdrawal}}$ arising whenever DFI is withdrawn from the CEX. A total of 246 withdrawals with an average withdrawal volume of 9118.18 DFI are withdrawn during the time horizon of the experiment. A total of 2,243,071 DFI is withdrawn at a cost of $97.54.



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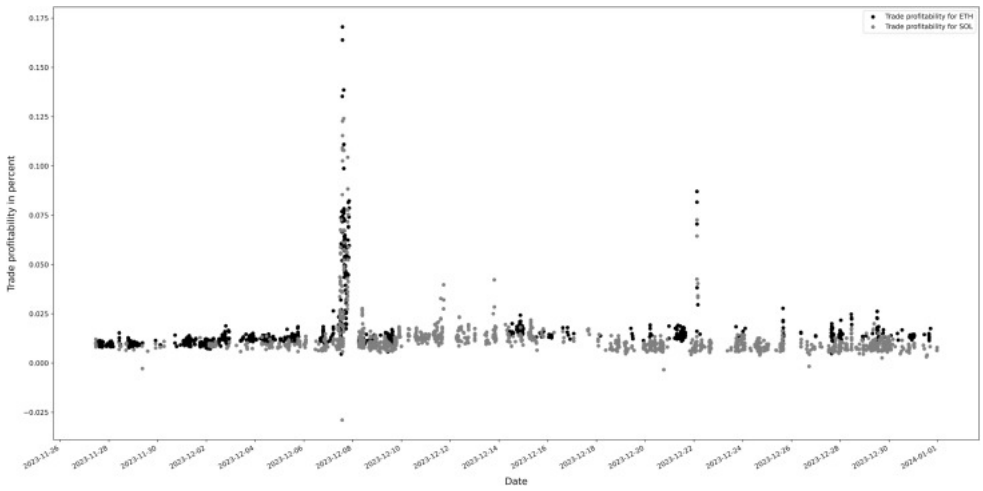
Fig. 7. Cumulative fees dMFP.

Further, the travel cost C_{travel} includes costs that occur through wrapping activities F_{wrapping}. The wrapping provider of the DeFiChain charges a variable fee of 25 basis points on the wrapped volume, along with a fixed fee to cover the Solana transaction cost $F_{\text{transaction}}$ for transferring funds from the wrapping provider to the CEX. A total of 4870.14 Solana is circulated from the blockchain through the wrapping provider towards the CEX. The average withdrawal volume is 143 Solana, at a total cost of $25.73. The associated total wrapping fees amount to $823.42, which is the highest cost factor after the trading fees. The initial portfolio value amounts to $21,565.5 on November 27, 2023. The total trading profit amounts to $6318.47, and the net profit is $5371.8. The return on investment between November 27, 2023, to December 31, 2023, amounts to 24.9 percent.

In the following, the paper analyzes the second arbitrage system dD6K, which combines Ethereum and Solana arbitrage bots. Initially, the goal was to operate the Ethereum and Solana bots independently; however, this approach was later adjusted to improve liquidity efficiency. The initial exposure portfolio for the dD6K is 50,000 DFI, 4200 USDT, 150 SOL, and 20.2 ETH. While two arbitrage bots of the dD6K operate independently, they share common liquidity in DFI and USDT exposure. To maintain a trade volume of $1500 for each strategy, the exposure position in DFI and USDT is increased by 25,000 DFI and 2100 USDT. This adjustment ensures that both arbitrage bots do not compete for liquidity simultaneously, thereby preventing potential interference when accessing DFI or USDT resources. As with the dMFP arbitrage system, we selected a trade volume of $1500 to keep the expected slippage low.

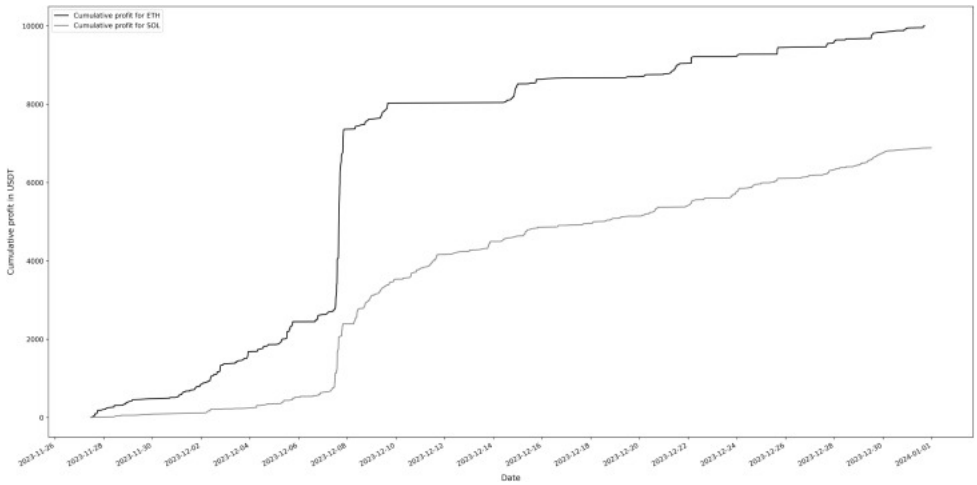
The Ethereum and Solana bots perform 996 and 1236 successful arbitrage trades, respectively. The average realized profit per trade is 1.24 percent ($5.57) for the Solana bot and 1.62 percent ($10.04) for the Ethereum bot. For all three bots in the two arbitrage systems, the average realized profit exceeds the expected realized profit, which is a result that arises through outlier trades. A total of four trades are unprofitable, leading to an overall loss of $5.85.

Fig. 8 illustrates the distribution of realized arbitrage profit in percent for both bots from the dD6K arbitrage system. As with the dMFP trading system, the realized profit incorporates the trading costs C but excludes the cost C_{travel} that occurs whenever liquidity rotates between the exchanges. The trading fees and configuration of both bots are the same as for the bot in the dMFP system, targeting an expected profit of 75 basis points. Fig. 9 shows the cumulative trade profit for the dD6K arbitrage system. It depicts the profit of the Ethereum and Solana bots that achieve a total trading profit of $9999.84 and $6884.52, respectively. For this, it required a total trading volume filled for bid and ask limit orders at the CEX that amounts to $1,060,672. The top 10 percent of all arbitrage trades account for 51.67 percent of the overall profit. Fig. 10 depicts the cumulative travel costs for the dD6K arbitrage system. The system combines two trading bots that have withdrawn liquidity from the CEX to the blockchain whenever DFI liquidity is low. A total of 323 withdrawals are executed at an average volume of 17,141 DFI. A total of $128.07 was paid to withdraw 5,536,804 DFI. The return on investment for the dD6K arbitrage system is 33.35 percent for the Solana bot and 17.39 percent for the Ethereum bot.



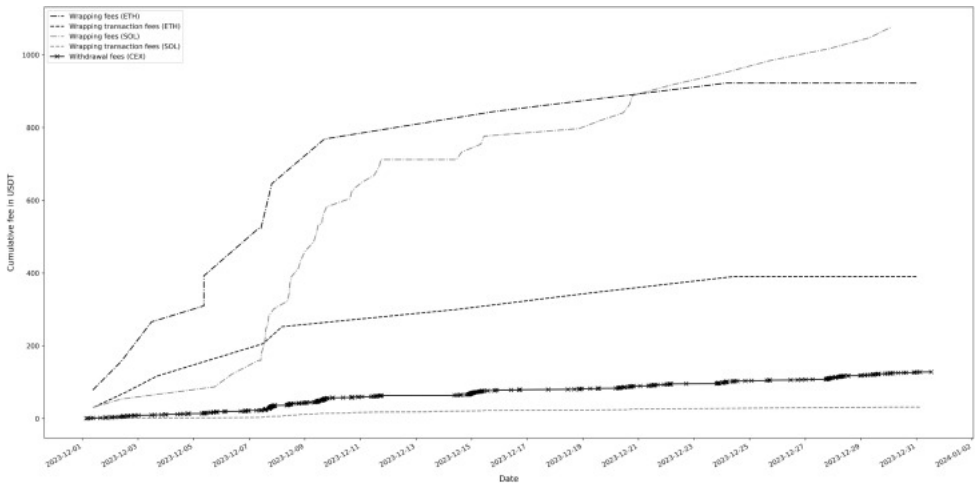
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Fig. 8. Trade profitability dD6K.



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Fig. 9. Cumulative profit dD6K.



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Fig. 10. Cumulative fees dD6K.

Table 3 summarizes the performance of both arbitrage systems. A comparison between the dMFP and dD6K arbitrage systems reveals the economies of scale advantage inherent in an arbitrage system that uses multiple bots. The average withdrawal volume is 17,141 DFI, unlike 9118.18 DFI for the dMFP arbitrage system. The withdrawal cost is a fixed cost, which allows the

dD6K arbitrage system to rotate DFI liquidity more cost-efficiently compared to the dMFP arbitrage system. Another advantage that results from the combined portfolio is an increased number of executions. The dMFP arbitrage system utilizes 225 Solana for arbitrage trading, while the dD6K arbitrage system utilizes 150 Solana. However, the dD6K system has 1236 successful arbitrage trades compared to the 1074 arbitrage trades of the dMFP system. The distribution of arbitrage trades made for both systems shows that multiple executions are centered on specific dates. In other words, if arbitrage opportunities occur, they often persist and allow multiple executions. The dD6K arbitrage strategy benefits during a price shock in one currency, as the shared liquidity of the DFI allows multiple executions until the bot exhausts its DFI liquidity. This explains the higher number of executions for the dD6K system compared to the dMFP system, which holds a higher exposure position in Solana. The total wrapping fees for the dD6K arbitrage system amount to $1077.11 and $31.41 for the wrapping and transaction fee for the Solana bot, and $922.3 and $390.4 for the Ethereum bot.

Table 3. Performance summary of arbitrage systems.

Empty Cell	**dMFP system - SOL**	**dD6K system - SOL**	**dD6K system - ETH**
Portfolio DFI	25,000 ($6150)	50,000 ($12,300)	x
Portfolio USDT	2100 ($2100)	4200 ($4200)	x
Portfolio ETH	x	x	20.2 ($41,694)
Portfolio SOL	225 ($13,316)	150 ($8877)	x
Total portfolio value	$21,565	$67,071	–
Number of trades	1074	1236	996
Average realized spread	1.54%	1.24%	1.62%
Average profit	$5.88	$5.57	$10.04
Total trading profit	$6318.47	$6884.52	$9999.84
CEX DFI withdrawals	246	323	–
CEX average volume DFI withdrawal	9118 DFI	17,141 DFI	–
Total withdrawn amount	2,243,071 DFI	5,536,804 DFI	–
Total cost	$97.54	$128.07	–
Total wrapping volume	4870.14 SOL	5794.71 SOL	164.56 ETH
Average withdrawal	143.24 SOL	137.97 SOL	18.28 ETH
Total transaction fee	$25.73	$31.41	$390.42
Total wrapping fee	$823.42	$1077.11	$922.31
ROI	24.91%	33.35%	17.39%
Net profit	$5371.78	$5711.97	$8687.14

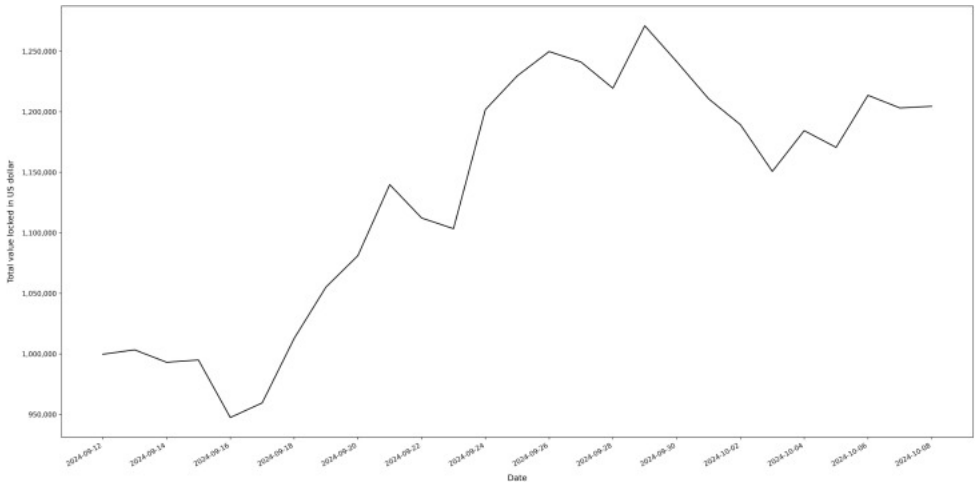
Note: This table compares the dMFP and dD6K arbitrage systems in their exposure position, trade activities, underlying costs and profits for the trading period from November 27, 2023 until December 31, 2023. The dMFP operates one arbitrage bot, whereas the dD6K operates two arbitrage bots within the arbitrage system. The dD6K system jointly shares the DFI and USDT liquidity for SOL and ETH trading bots.

The main reasons for the high number of profits for both arbitrage systems are high TVL levels, low competition and high price divergence of DFI to SOL and ETH between November 27, and December 31, 2023. While DFI depreciated from $0.246 to $0.159, Solana and Ethereum appreciated from $59.18 to $101.99 and $2064.07 to $2294.34, respectively. The static price of the AMM protocol allowed arbitrage traders to continuously benefit from price discrepancies between the DEX and CEX. As the price divergence comes to the advantage of arbitrage profits, it comes with a loss in the exposure position. The exposure position for dD6K, which has 50,000 DFI, results in a total loss of $4350.

5. Robustness check

Following the successful deployment of the arbitrage trading system on DeFiChain, our next objective is to evaluate the robustness of the algorithm by testing its adaptability across blockchains with different conditions. To select a suitable blockchain for this testing phase, we prioritize networks with low transaction fees, the presence of an AMM smart contract with at least $0.5 million in TVL, and the ability to facilitate the circulation of assets between exchanges, thereby allowing arbitrage conditions. This environment differs from DeFiChain in that it combines lower trading costs with generally lower liquidity levels. We identify the SEI blockchain as a suitable testing ground. Unlike the initial arbitrage system deployed on DeFiChain (a Bitcoin fork with a proof-of-stake consensus mechanism) the second system is implemented on SEI, an EVM-compatible, high-performance Layer 1 blockchain optimized for transaction speed.

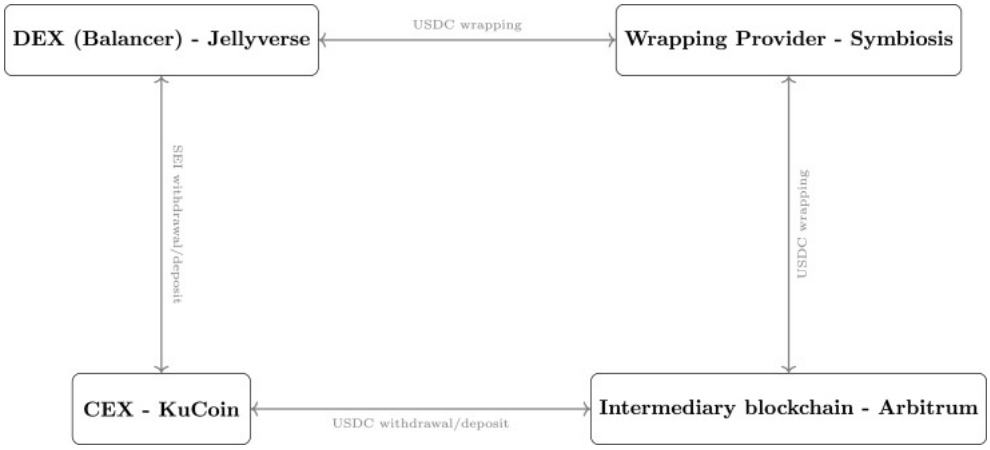
On the SEI blockchain, we select the Jellyverse DEX and implement a two-point arbitrage strategy using the SEI-USDC trading pair. For simplicity, the corresponding CEX trade is executed on KuCoin using the SEI-USDT pair.[18] Fig. 11 shows the TVL for the SEI-USDC trading pair during the system's operating period from September 12, 2024, to October 8, 2024, starting at $1 million and increasing to $1.2 million.



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Fig. 11. Total value locked on SEI.

The block time of the SEI blockchain is approximately 400 ms (0.4 s), with a throughput of up to 12,500 transactions per second.[19] SeiScan[20] shows historical transaction fees that are consistently below $0.05, aligning well with the algorithm's requirements. Jellyverse,[21] an AMM protocol based on the Balancer model, offers stable and weighted pools with sufficient liquidity, ensuring that the bot can operate with sufficient trading volumes.[22] Symbiosis operates as a bridge provider to circulate USDC funds into the SEI blockchain.[23]
As Fig. 12 shows, USDC liquidity can be withdrawn from Kucoin to the Arbitrum blockchain and subsequently bridged to the SEI blockchain. Kucoin remains the CEX as it allows the algorithm to benefit from hidden orders. For trading at Jellyverse and Kucoin, the third trading system 0x2f uses the trading pairs SEI-USDC and SEI-USDT, respectively.[24] To circulate USDC through Symbiosis from Kucoin to the SEI blockchain, we use the CEX trading pair USDC-USDT to convert USDT to USDC. Unlike the triangular arbitrage strategy used on the DeFiChain, we implement a two-point arbitrage strategy on the SEI blockchain, assuming USDT and USDC maintain a stable parity value. The execution of two-point arbitrage has lower trading fees, as it requires only two trades rather than three. Furthermore, the limit order fee on KuCoin for the SEI market is 8 basis points, compared to 16 basis points for DFI. Given the bot's reduced transaction costs, the spread threshold ρ is reduced from 120 basis points on DeFiChain to 75 basis points on SEI to enhance execution frequency.

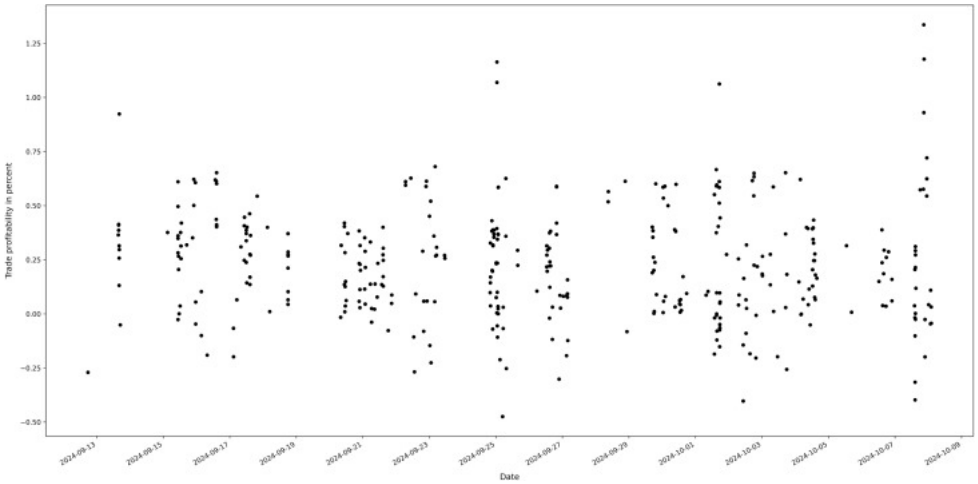


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Fig. 12. SEI and USDC flow.

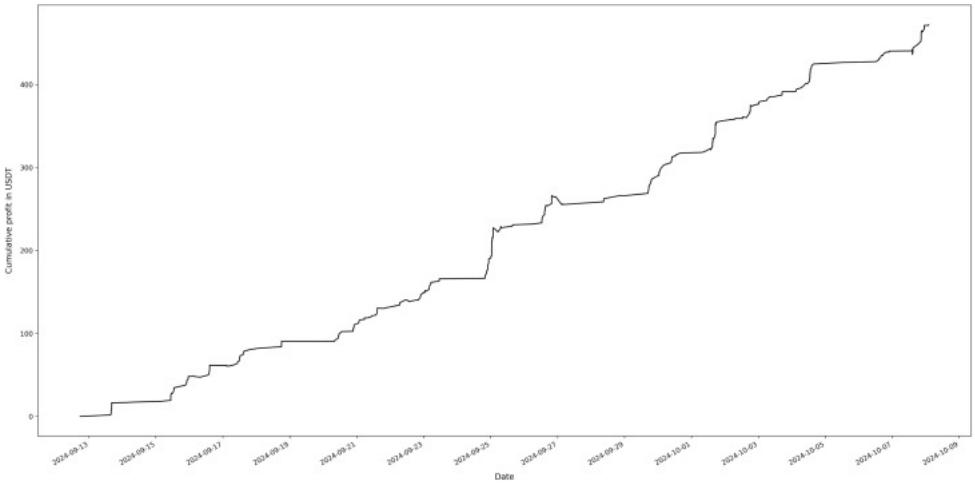
The comparably low TVL in the smart contract forces the configuration to operate under a reduced trading volume of $600. This is because the configuration aims to keep the expected slippage low to guarantee profitable arbitrage trades. The 0x2f system uses an exposure position of 12,000 SEI, 2100 USDT, and 2100 USDC. At the start of the trading period, prices were: SEI $0.282, USDT $1, and USDC $1. A trade on the Balancer model, Jellyverse, for the SEI-USDC trading pair charges a fee of $f\text{DEX}=30$ basis points, the limit order fee amounts to $f\text{CEX}_{\text{limit}}=8$ basis points and the spread level is configured as $\rho=75$ basis points. For a trading volume of $600, the expected slippage at the DEX is $s\text{DEX}=[10,12]$ basis points. Correspondingly, the expected profit for the 0x2f system is 25–27 basis points.
Fig. 13 depicts the distribution of realized arbitrage trades for the 0x2f system. The average realized arbitrage spread is 23 basis points which aligns with the expected value. Out of 411 trades, 319 trades are profitable, and 92 are unprofitable, with the total loss from all unprofitable trades amounting to $47.77. Frequent minor negative trades occur because, instead of wrapping USDC and USDT through the wrapping provider, we aim for autorebalancing trade. As discussed in Section 2.2, we apply autorebalancing trades to reduce/mitigate travel costs. To avoid the circulation of funds, which are connected with comparably high costs C_{travel}, autorebalancing allows the reduction of the spread level for either order (bid or ask). If the bot runs out of

liquidity on one position, autorebalancing reduces the spread level. Whereas the design of the bot targets a profit of around 25–27 basis points, “autorebalancing = 30 basis points” further reduces the profit for such trades to around 0 basis points.



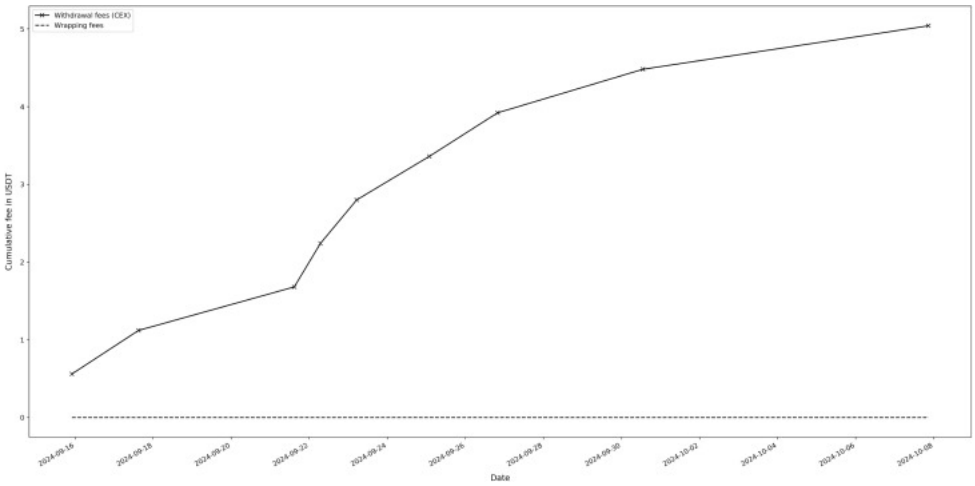
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Fig. 13. Trade profitability 0x2f.



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Fig. 14. Cumulative profit 0x2f.



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Fig. 15. Cumulative fees 0x2f.

Fig. 15 depicts the travel fees C_{travel} that arise through the circulation of capital between the two exchanges. A total of 9 withdrawals with an average volume of 5444 SEI are withdrawn from the CEX to the SEI blockchain, incurring a total withdrawal cost of $F_{\text{withdrawal}}$=$5.04. The 0x2f trading system rebalanced its exposure position via autorebalancing, eliminating the need for USDC or USDT circulation and thereby avoiding any wrapping fees F_{wrapping}=$0. However, the avoidance of wrapping fees comes at the cost of negative trades amounting to $47.77. Fig. 14 shows the cumulative profit over the 26-day period for the 0x2f trading system, demonstrating a total profit of $471.73 and confirming the algorithm's robust performance on a new blockchain with low TVL value in the trading pair. The net profit of the 0x2f trading system is $466.70, yielding a return on investment of 6.15 percent.

6. Results discussion

The deployment of the arbitrage system on the SEI blockchain demonstrates that our algorithm remains effective in executing two-point arbitrage on newer blockchain platforms optimized for high-speed transactions. A comparative performance analysis of the two arbitrage systems operating on the DeFiChain and SEI blockchains shows that the system on DeFiChain achieves significantly higher profitability. This outcome aligns with our ex-ante expectations, as DeFiChain offers higher spreads and larger TVL which allow the execution of larger arbitrage volumes. The average realized arbitrage spread on the SEI system is 23 basis points compared to 124, 154 and 162 basis points on the DeFichain. Both DEXs on the DeFiChain and SEI blockchains are not integrated into any DEX aggregator, resulting in fragmented liquidity that makes them attractive markets for arbitrage. The two DEXs differ in their competitiveness, liquidity level, trading and wrapping costs, which explains differences in the performance of the arbitrage bot across the two environments.

The inherent complexity of the triangular arbitrage strategy on DeFiChain, compared to the simpler two-point strategy on the SEI blockchain, may explain the lower level of competition. Lower competition enables the execution of larger arbitrage volumes. The trust in the wrapping provider can also give reason for reduced arbitrage competition. With increased disbelief in the wrapping provider, arbitrage competition may be reduced or trader may widen their required spread to compensate their risk.

An additional factor contributing to the higher arbitrage profits on DeFiChain is the magnitude of spot price movements. During the operation period of the arbitrage systems, SOL and ETH appreciated by 198.38 percent and 80.40 percent against DFI, respectively, while SEI increased by 48.30 percent against USDC. Larger spot price changes allow for larger arbitrage volumes. The combination of high TVL, significant spot price volatility, and low market competition creates an ideal environment for arbitrage traders.

The lucrative profits achieved with our comparable mid-market capitalization trading pairs SEI-USDT, DFI-SOL, and DFI-ETH raise the question whether similar profits can be earned with major trading pairs such as ETH-USDT or BTC-USDT. According to data from CoinMarketCap and CoinGecko, there are more than 500 decentralized exchanges (DEXs) and approximately 200–250 centralized exchanges (CEXs) where BTC and ETH can be traded. Uniswap V4 alone holds significant liquidity, with a total value locked (TVL) of 166.4 million USD in a single ETH-USDT pool and 120 million USD in a single WBTC-USDC pool. For DEX to CEX arbitrage higher TVL values on Uniswap enable larger arbitrage volumes and reduce the slippage costs faced by arbitrage traders. Conversely, arbitrage competition in these highly liquid environments is considerably greater, leading to smaller spreads at which arbitrage opportunities are executed. For example, arbitrage systems connecting Uniswap V4 on Ethereum with Binance or KuCoin are unlikely to yield profits for individual arbitrage traders. Companies that specialize on arbitrage trading dominate in these markets due to economies of scale, as they benefit from negative fee structures on CEXs and faster execution on DEXs through prioritized transactions. Unfortunately, these companies are reluctant to share their executed trade data with researchers, as doing so could jeopardize their business activities. Historical data indicates that cross-exchange arbitrage trades between Uniswap and Binance rarely exceed 10 basis points, which is too narrow to generate sufficient profits for our arbitrage system. This shows while major pairs such as ETH-USDT and BTC-USDT exhibit much deeper liquidity and lower slippage, the intense competition in these markets compresses spreads that preclude consistent arbitrage profits for smaller participants. Therefore, this study focuses on mid-market pairs that operate in less efficient markets where liquidity is sufficient and spreads are wide enough to execute profitable trades. Hence, our results are representative of DEX to CEX arbitrage conditions in semi liquid markets rather than hyper efficient asset environments. Given the large number of DEX and CEX trading pairs available, numerous combinations of trading pairs exist that can yield promising returns. The spreads depend on the underlying cost structure, while slippage varies with liquidity levels on the respective DEX.

Table 4 provides an overview of arbitrage environments with favorable characteristics, namely blockchains with low transaction fees, DEXs with sufficient liquidity, and CEXs that allow to mimic trading pairs to effectively balance arbitrage positions. The table shows DEX to CEX combinations for two-point and triangular arbitrage. For our proposed algorithm we believe that triangular arbitrage strategies tend to be more lucrative due to less competition. When setting up an own arbitrage system user face several risks, including not being able to hedge exposure of assets, the possibility of frontrunning attacks, competition from other arbitrageurs, liquidity provision attacks, all of which can lead to unprofitable trades. Reliance on wrapped assets introduces another risk, potentially resulting in total loss if the issuer fails. Therefore, a thorough understanding of the risk and cost structure is essential for selecting an appropriate spread that ensures systematic profits. Concluding, while arbitrage remains a profitable strategy with lucrative returns, its success strongly depends on the conditions of the environment, i.e., the blockchain, trading pair, CEX, DEX and the wrapping provider.

Table 4. Overview of arbitrage environments.

DEX	DEX pair	TVL	CEX	CEX pair(s)
Meteora	TRUMP–USDC	$305M	Binance	TRUMP–USDC
Raydium	MEW–SOL	$25M	HTX	MEW–USDT, SOL–USDT
Raydium	SOL–Fartcoin	$15M	KuCoin	FARTCOIN–USDT, SOL–USDT
Raydium	WIF–SOL	$12M	KuCoin	WIF–USDT, SOL–USDT
Raydium	RAY–USDC	$6M	Binance	RAY–USDC
Momentum	WAL–SUI	$3M	Binance	WAL–USDT, SUI–USDT
PancakeSwap	MERL–USDC	$2M	Gate	MERL–USDT, USDT–USDC
PancakeSwap	SOON–WBNB	$1.4M	Gate	SOON–USDT, BNB–USDT
Uniswap	PEPE–ETH	$49.5M	KuCoin	PEPE–USDT, ETH–USDT

DEX	DEX pair	TVL	CEX	CEX pair(s)
Uniswap	FLOKI–ETH	$16.6M	Binance	FLOKI–USDT, ETH–USDT
Hyperion	APT–USDC	$8.2M	MEXC	APT–USDT, USDC–USDT
Stellar X	XLM–USDC	$5.7M	HTX	XLM–USDT, USDT–USDC
Trader Joe	AVAX–USDC	$12.4M	KuCoin	AVAX–USDC
Ston.fi	STORM–TON	$1.1M	MEXC	STORM–USDT, USDT–TON

7. Outlook

The ecosystem of DeFi evolves rapidly. DEXs continue to introduce new functionalities to expand the toolkit available to both liquidity providers and traders. The evolution from Uniswap V2 to V4 reflects a systematic attempt to address growing inefficiencies for traders, liquidity providers, order routing, and transaction costs. Each new version responds to specific frictions observed in on-chain trading. The innovations enhance capital efficiency and reduce operational costs for traders. However, they also impact arbitrage traders.

Under the Uniswap V2 design, liquidity providers were required to allocate capital uniformly across the entire price spectrum $[0,\infty]$ resulting in a substantial portion of liquidity being deployed in price ranges far from the spot price. This standardization of capital allocation leads to low capital utilization for liquidity providers. Arbitrage traders profit from the inflexible allocation of capital within the Uniswap V2 version, as they consistently profit from stale capital, whenever the market price changes. Uniswap identified that traders (and arbitrageurs) experience comparable high slippage cost in the V2 version, due to insufficient amount of capital allocated around the spot price. To counteract the inefficiency for traders, the introduction of concentrated liquidity in the Uniswap V3 allows liquidity providers to confine liquidity to specific price intervals (“ticks”), which increases capital available for traders (and arbitrageurs) and improves capital utilization and fee income for liquidity providers.

The narrative for liquidity provision changed, from long-term, inflexible liquidity provision (“deposit and forget”) towards monitoring and managing liquidity positions more actively within liquidity ranges. To the disadvantage for arbitrage traders, the flexibility of capital allocation has reduced the amount of stale capital that arbitrage traders used to profit from. To the advantage for traders (and arbitrageurs), the concentration of liquidity within specific tick ranges effectively reduces price slippage, thereby lowering trading costs. However, note that, the tick structure can also fragment liquidity across tick ranges that leads to high slippage costs. For arbitrage traders, this means that slippage calculations become more complicated and compute intensive, as trades executed within active tick ranges experience deep liquidity at minimal slippage, while trades outside those ranges can face steep slippage. Consequently, comparable to limit order book systems the algorithm of arbitrage traders needs to continuously update slippage calculation and tick-distribution to adjust trading volumes which allow to realize profitable spreads. The evolution of DEX innovations, underscores a shift from comparable predictable arbitrage towards precision-based, data intensive arbitrage.

The Uniswap V4 model enhances decentralized trading, reducing frictions for trading and advancing flexibility for liquidity providers. This model introduces two key architectural innovations, the singleton design and flash accounting. In the Uniswap V4, all liquidity pools reside within a single “singleton” contract that both facilitates pool creation and stores the state of all pools. Unlike the Uniswap V3, where each pool was a separate contract deployed via a factory, the V4’s design eliminates the gas intensive process of deploying a new contract for every pair. This drastically lowers the cost of pool creation and enables more efficient multi-hop swaps because all pool states are stored within a single system. For arbitrageurs and traders, this means that complex routing paths across multiple pools can be executed faster and more cheaply. Additionally, the Uniswap V4 introduces flash accounting a mechanism allowing all balance-changing operations, e.g., multiple swaps or liquidity adjustments, to be executed within one atomic transaction, with final token transfers settled at the end. In the Uniswap V3, each swap between pools required individual token transfers, which increased gas costs and latency for traders and arbitrage traders. By deferring settlement until after all operations, flash accounting enables more efficient multi-hop as for triangular arbitrage strategies. The addition of programmable hooks in the V4 improves the flexibility of the model for liquidity providers and traders, but adds complexity for arbitrage traders. Hooks modify conservation functions, which effects slippage calculation, fee levels, and spot prices that are now customizable and depend on the corresponding hook. This unpredictability complicates liquidity estimation and spot price calculation for arbitrageurs.

In contrast, traders benefit from hooks, as it strengthens interoperability between protocols, reduces idle capital, and allows capital to be accessible for traders once it is needed. Overall, while the Uniswap V4’s architecture through the singleton model, flash Accounting, and hooks lowers structural inefficiencies and transaction costs, it also increases the technical sophistication required to identify and exploit arbitrage opportunities. As these mechanisms make markets more complex, integrated and less fragmented, the remaining arbitrage opportunities are likely to cluster among well-capitalized, technologically advanced participants. This reflects a broader tendency in financial markets: as complexity, interoperability and automation rise, arbitrage activity becomes increasingly centralized among a few specialized actors (Schleifer and Vishny, 1997) while trader and liquidity provider benefit from enhanced simplification, flexibility and automation.

To sum up, DeFi innovations such as DEX aggregators and new order flow solutions will further reduce arbitrage opportunities between DEXs (DEX to DEX) within the same blockchain. DEX to DEX with both DEXs on different blockchains may remain a lucrative strategy. Arbitrage opportunities between CEXs (CEX to CEX) are currently dominated by a small number of market makers who benefit from economies of scale and preferential fee structures. On the contrary, DEX to CEX arbitrage opportunities are likely to persist and can be effectively captured using arbitrage systems such as those documented in this paper. Arbitrage traders who perform DEX to CEX arbitrage and place quotes in CEX order books rely on certainty regarding spot prices, fee levels, and slippage at the DEX. Any uncertainty in these factors leads to increased risk of the arbitrage system. As DeFi continues to evolve and consensus mechanisms change, arbitrageur traders must continually adapt their systems to stay competitive. While DeFi follows the narrative to mitigate intermediates and third parties, arbitrage traders themselves act as intermediaries. They facilitate capital flows and expect a compensation for the associated risk and service. This intermediation introduces financial friction which reduces the trade efficiency.

In the absence of a direct order flow solution between DEXs and CEXs, frequent DEX to CEX arbitrage opportunities are likely to persist. To improve price efficiency and to reduce liquidity fragmentation within the cryptocurrency ecosystem, CEXs should find solutions to directly connect with DEX liquidity. To reduce frictions in the DeFi ecosystem, DEX that operates on different blockchains should connect their capital through direct order flow solutions.

8. Conclusion

The paper demonstrates the lucrativeness of arbitrage trading between decentralized and centralized cryptocurrency exchanges. Three independent arbitrage trading systems earned a total profit of 19,770.78 USDT for the time horizon from November 27, 2023, to December 31, 2023, and $466.70 from September 12, 2024, to October 08, 2024. For all three arbitrage trading systems, the realized arbitrage spreads align with or exceed the expected spreads.
Practical arbitrage differentiates from theoretical back-testing of arbitrage opportunities as it includes all operational costs, liquidity constraints, and direct effects on the markets. Therefore, our economic model includes all operational costs and frictions arbitrage traders exhibit at praxis and documents them empirically. Further, it discusses the trade-offs for various cost adjustments to maximize the profit.
We demonstrate various issues when interacting with smart contracts that need to be studied before setting up an arbitrage system. Unlike high-frequency trading at CEXs, trading at the DEX is not purely dependent on fast executions. We disclose our trading bot's algorithm, enabling readers to replicate it on blockchains with similar characteristics. While blockchain innovations, such as DEX aggregators, have enhanced the efficiency of order flow for on-chain activities, there remains significant potential to improve the order flow between DEXs and CEXs to reduce cross-exchange spreads and optimize market efficiency. Smart contract-based solutions that connect both exchanges, as proposed by Jansen (2023), could help to overcome the inefficiency of fragmented capital in the cryptocurrency markets.

CRediT authorship contribution statement

Lennart Schwertfeger: Writing – original draft, Software, Investigation, Data curation. **Bodo Vogt:** Writing – review & editing, Resources, Formal analysis, Conceptualization.

Appendix. Additional information

Xu et al. (2023) present a comprehensive analysis of various AMM smart contracts. They categorize contracts based on their conservation functions, i.e., the mathematical pricing model behind AMMs, asset provisioning mechanisms, exchange rate determination, and slippage calculations. It requires various conservation functions to allow efficient trading for specific trading pairs. For example, Balancer,[25] originally launched on the Ethereum[26] blockchain, offers two specific trading pools: weighted pools and composable stable pools. Weighted pools are designed for trading assets with volatile prices. In contrast, stable pools are optimized for trading assets that maintain stable prices, such as fiat currencies, e.g., USDC and USDT, which aim to peg to the value of the U.S. dollar.
Despite the adjustment of conservation functions towards specific trading pool needs, other DEX innovations, such as smart order routing, have improved the efficiency of DEX trading by automating price comparisons among trading pools. Before these innovations, DEX traders had to manually compare prices across different trading pools to achieve the best possible execution price. Smart order routing selects the most efficient route for trades and can also split trades across multiple routes to further enhance price efficiency. While these innovations improved trading on individual exchanges, DEX aggregators have expanded this optimization by improving order flow across multiple exchanges on a single blockchain. Similar to smart order routing, DEX aggregators like Jupiter[27] (on the Solana[28] blockchain) and 1inch[29] (on the Ethereum blockchain) compare prices across various exchanges and liquidity pools to optimize order flow for traders.
The improved order flows for blockchain trading activities lead to reduced price differences along exchanges and trading pairs. The bypassing effect is a reduced amount of arbitrage opportunities for arbitrage traders. Arbitrage traders benefit from selling and buying an equivalent amount of an asset at the same time, as they take advantage of price differences. Arbitrage trader that interacts with AMM DEX, benefit from the static nature of the protocol. If the market price of an asset moves, arbitrage traders interact with the DEX to benefit from the old market prices. This happens at the cost of liquidity providers that fund the liquidity pool. Whereas liquidity providers profit from trading fees that arise through trading activities, they can also suffer a financial loss, so-called divergence loss (impermanent loss).
Divergence loss is the loss that results from changes in the exposure position of the liquidity provider. Through trading activities, the pool reserves and, correspondingly, the exposure position for liquidity providers change. The divergence loss effect leads to an increased exposure position of the under-performing and a decreased position of the over-performing asset. Loesch et al. (2021) investigate the behavior of liquidity providers at the Uniswap[30] DEX for the time horizon from May 5, 2021, to September 20, 2021. Their analysis shows that liquidity providers generated $199.3 million in fees while they lost $260.1 million due to the divergence loss of their funded assets. As they state, 49.5 percent of all liquidity providers had negative returns during the analyzed time horizon. Fritsch and Canidio (2024) use price data from Uniswap and Binance to simulate the amount of arbitrage opportunities. They find that losses of liquidity provider exceed fees earned, where Uniswap v2 users are better off compared to Uniswap V3 liquidity provider. In their analysis they use historical datasets which neglect transactions cost and assume that trades are executed at the best bid or ask price.

Data availability

Data will be made available on request.

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1
See official website.

2
See Appendix for additional information on AMM models and DEX aggregators.

3
See Xu et al. (2023) p. 238.

4
In Section 5, we deploy our arbitrage trading system on the Balancer model. Here, the constant product AMM conservation function includes a weighting of reserves.

5
There is a discrepancy between how slippage is understood in academic literature and how it is used in practice. In this paper, we follow the definition established in the academic literature. However, it is important to note that smart contract developers typically define price impact as the difference between the spot price and the realized price. Slippage, in contrast, includes not only the price impact of the trader's own transaction but also the impacts of other transactions executed beforehand. Accordingly, the slippage threshold set by the trader serves as a safeguard, ensuring that the actual trade output does not fall below the defined minimum.

6
See KCS.Foundation (2022).

7
This assumes that the trading bot utilizes a market order, which is subject to slippage.

8
See official website.

9
Depending on the wrapping provider, the fee is fixed or variable based on volume.

10
This can vary depending on the blockchain.

11
The mempool calculation is based on the Uniswap V2. However, to apply this calculation to other DEXs, it must be adjusted to their underlying conservation function.

12
See Baum et al. (2022) p. 5.

13
The same simulation can be replicated with different levels of reserves and coins that lead to identical results.

14
See official website.

15
To our knowledge, this client can only be used with Microsoft Windows operating systems.

16
The wallet address for dMFP system is dMa2PMnAudfZds8mnRm8EtjDWf7UWAuQFP, and the wallet address for the dD6K system is dDXmMofYrtdiqB4MuGewDEA6jJN8Yo4F6K. More information can be found on the DeFi Scan (see official website).

17
Trading fees at the CEX are reduced from 20 basis points to 16 basis points for the limit order and from 10 basis points to 8 basis points for the market order, due to a 20 percent discount applied for holding KCS.

18

Note that since the two trading pairs are denominated in different US dollars stablecoins, a USDC-USDT conversion is added at the CEX.

[19] See official website↗.

[20] See official website↗.

[21] See official website↗.

[22] See official website↗.

[23] See official website↗.

[24] The wallet address is 0x98890e37bfE73ED58944706076E37318769EB52f.

[25] See Martinell and Mushegian (2019).

[26] See Buterin (2014).

[27] See JupiterChain (2018).

[28] See Yakovenko (2017).

[29] See Bukov et al. (2024).

[30] See Uniswap (2020).

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